Conformalized Heteroscedastic Variant Effect Prediction with Local Coverage Guarantees

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Abstract

How fine should the Mondrian partition of a conformal predictor be under heteroscedasticity and class imbalance? The bin count K governs whether prediction sets attain bin-local coverage — the per-region calibration that pointwise conditioning provably cannot deliver in finite samples [Barber et al., 2020]. For binary conformal classification *within the equi-bin Mondrian-K family*, we prove a tight finite-sample $O(n^{-1/2})$ rate on the worst-cell coverage gap, $G(K^\star) \leq 2\sqrt{L_F R/(\pi_{\min} n)}$ at $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$, with a matching lower bound on the same family that exposes an explicit $\pi_{\min}^{-1/2}$ constant (Theorem 3, 4); the rate is dimension-free because conditioning is on the 1-D summary $\hat{\sigma}(x)$. The result instantiates as **HCCP** (Heteroscedastic Class-Conditional Conformal Prediction): a post-hoc framework pairing a learned variance head $\hat{\sigma}$ with Mondrian-$(y \times \hat{\sigma}$-bin$)$ calibration, plus a non-vacuous robustness corollary (T3$'$, via Barber et al. [2023]) for cell-level exchangeability violations detected by our per-chromosome KS audit. On TraitGym [Benegas et al., 2025], at the recommended operating point ($K_{\text{eval}} = 3$ Mendelian, 5 Complex; per-cell minority ≥ 100), HCCP is the only single-fold partition we evaluated — across split CP, $\hat{\sigma}$-Mondrian [Boström and Johansson, 2020], class-Mondrian, RLCP, weighted CP and SC-CP at matched $(\hat{p}, \hat{\sigma})$ — that holds $\text{cov}_{|Y=1} \geq 0.85$ and $\hat{\sigma}$-bin gap ≤ 0.20 on both Mendelian ($n = 3{,}380$) and Complex ($n = 11{,}400$). All baselines carry $B = 200$ chromosome-level bootstrap CIs; the same HCCP calibration on the published LogReg aggregator gives essentially identical $\hat{\sigma}$-bin gap, decoupling the conformal-side gain from base-predictor accuracy. A K_{eval} sweep (App. C.1) shows HCCP's Complex advantage is robust across $K_{\text{eval}} \in [2, 10]$; on Mendelian weighted CP narrows the margin to $1.15\times$ at the recommended operating point and overtakes HCCP for $K_{\text{eval}} \geq 5$ (per-cell minority < 70). Cross-domain replication on ProteinGym [Notin et al., 2023]. Separately, our GBM aggregator reaches Mendelian AUPRC 0.900 ($+14.8$ pp over the published LogReg baseline).

1 Introduction

Clinical variant interpretation drives consequential decisions [Richards et al., 2015], yet machine-learned variant effect predictors (VEPs) collapse evidence into a single probability [Kircher et al., 2014, Linder et al., 2025, Benegas et al., 2023, 2025] without an explicit notion of confidence. Conformal prediction provides distribution-free coverage guarantees [Vovk et al., 2005, 2003, Romano et al., 2019, Barber et al., 2021, 2023], but Barber et al. [2020] rule out pointwise conditional coverage in finite samples. What is missing for VEP is a composition that (i) consumes a learned heteroscedastic

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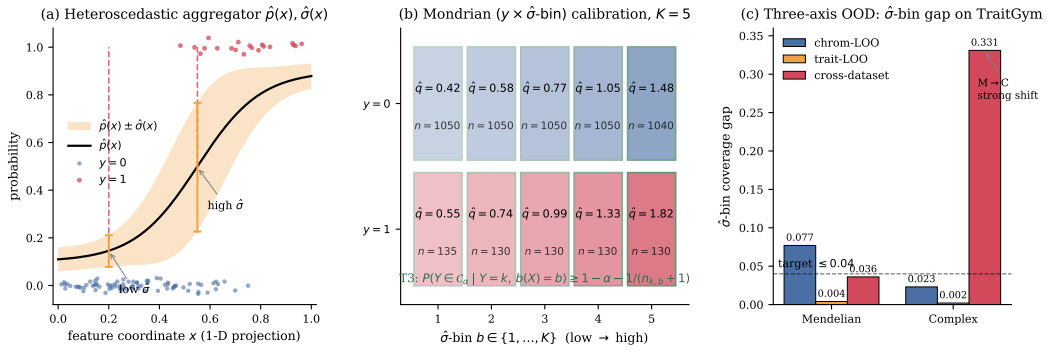

Figure 1: HCCP at a glance. **(a)** Heteroscedastic aggregator outputs $(\hat{p}, \hat{\sigma})$; nonconformity score $s(x, y) = |y - \hat{p}(x)|/\hat{\sigma}(x)$ scales the decision radius by local uncertainty. **(b)** Mondrian calibration partitions the labeled data by $(y \times \hat{\sigma}\text{-bin})$; per-cell quantile thresholds $\hat{q}_{k,b}$ restore class-conditional (T2) and bin-conditional (T3) coverage, while remaining compatible with Barber et al. [2020]'s pointwise impossibility. **(c)** Three-axis OOD validation on TraitGym: trait-LOO reaches the finite-sample floor 0.002–0.004; chrom-LOO and C→M cross-dataset stay ≤ 0.077.

head $\hat{\sigma}(x)$ as in DEGU [Zhou et al., 2026], (ii) stratifies calibration by both label and $\hat{\sigma}$-bin to regain as much conditionality as finite samples allow, and (iii) carries local coverage guarantees across distribution shifts. The gap is binding for VEP: TraitGym minority-class prevalence is $\pi_{\min} \approx 0.10$, and chrom-LOO + trait-LOO + cross-dataset shifts are intrinsic.

The technical question is: which Mondrian partition simultaneously delivers (i) class-conditional coverage under extreme imbalance and (ii) bin-local coverage under heteroscedasticity? Single-axis answers fail individually — $\hat{\sigma}$-Mondrian [Boström and Johansson, 2020] attains local coverage in shape but loses minority-class coverage ($\mathrm{cov}_{|Y=1} = 0.62$ on TraitGym Complex; the resulting minority-cell undercoverage drives its $\hat{\sigma}$-bin gap up to 0.83); class-Mondrian [Sadinle et al., 2019] attains class coverage but leaves a bin-local gap $13.4\times$ wider than the joint partition on Complex (bootstrap mean, $B = 200$). Within the equi-bin Mondrian-K family, we prove the joint $(y \times \hat{\sigma}\text{-bin})$ Mondrian attains both with worst-cell coverage gap $G(K^\star) \leq 2\sqrt{L_F R/(\pi_{\min} n)}$ (Theorem 3) and a matching $\Omega(\sqrt{L_F R/(\pi_{\min} n)})$ finite-sample lower bound (Theorem 4); $\pi_{\min}$ enters *explicitly* as a $\pi_{\min}^{-1/2}$ constant.

Contributions.

1. **Tight finite-sample rate within the equi-bin Mondrian-K class with $\pi_{\min}$ explicit in the constant** (Theorems 3, 4). To our knowledge this is the first finite-sample bound for class-conditional + heteroscedastic-local CP in which the minority-class probability $\pi_{\min}$ appears in the rate; the constant is dimension-free under a 1-D Lipschitz assumption on $\hat{\sigma}$. Concurrent imbalanced-CP work [Xie et al., 2025, Ding et al., 2023, Shi et al., 2024] provides marginal or class-cond rates without the $\pi_{\min}$-local product.

2. **HCCP, a post-hoc framework with an exchangeability-robust corollary.** A nonconformity score $s(x, y) = |y - \hat{p}(x)|/\hat{\sigma}(x)$ paired with Mondrian-$(y \times \hat{\sigma}\text{-bin})$ calibration. T3 gives exact bin-conditional coverage at rate $1/(n_{k,b}+1)$; T3′ (via Barber et al. [2023] Thm. 2 at the cell level) supplies a uniform finite-sample lower bound when our per-chromosome KS audit detects cell-level exchangeability violations on Mendelian.

3. **Empirical validation on TraitGym + ProteinGym + synthetic.** On TraitGym Mendelian / Complex at $\pi_{\min} = 0.10$, at the recommended operating point ($K_{\text{eval}} = 3$ / 5; per-cell minority ≥ 100), HCCP is the only single-fold partition we evaluated — across split CP, $\hat{\sigma}$-Mondrian, class-Mondrian, RLCP, weighted CP, and SC-CP at matched $(\hat{p}, \hat{\sigma})$ — that holds $\mathrm{cov}_{|Y=1} \geq 0.85$ and $\hat{\sigma}$-bin gap ≤ 0.20 on both datasets ($B = 200$ chrom-bootstrap CIs throughout). The conformal gain is decoupled from base-predictor accuracy (App. C.8).

ProteinGym [Notin et al., 2023] cross-domain replication; synthetic n-sweep verifies the dimension-free $O(n^{-1/2})$ rate.

2 Related work

Conformal prediction with conditioning. Split / Mondrian CP [Vovk et al., 2005, 2003, Angelopoulos and Bates, 2022] provides marginal / group-conditional coverage; Barber et al. [2020] rule out pointwise conditional coverage. Normalized scores with a learned $\hat{\sigma}$ originate with Papadopoulos et al. [2008] and Lei et al. [2018] for regression; Boström and Johansson [2020] systematized Mondrian conformal regression with normalized scores; CQR [Romano et al., 2019] is the regression analog of our score. Recent local-CP work targets feature-space coverage at slower rates: **RLCP** [Hore and Barber, 2025] achieves $O(n^{-2/(d+2)})$ kernel-localized coverage (intractable at $d \approx 7{,}700$); **Self-Calibrating CP** [van der Laan, 2024] achieves $O(n^{-2/3})$ on the $\hat{\mu}$-axis via isotonic monotonicity [Deng et al., 2021], on a different axis from ours. Tibshirani et al. [2019], Gibbs and Candès [2021], Barber et al. [2023] cover distribution-shift robustness; we use Barber 2023 Thm. 2 for our T3′. De Wolf et al. [2025] analyze conditional validity of heteroscedastic CP for regression; we extend to classification with $\pi_{\min}$ explicit. Toccaceli [2026] (concurrent) identifies optimal bin count under covariate-sorted contiguous binning as an open problem; Theorem 3 resolves a special case for ($y \times \hat{\sigma}$-bin) Mondrian under class imbalance.

Class-conditional and imbalanced CP. Sadinle et al. [2019] introduced label-conditional CP (Mondrian-by-y); Shi et al. [2024], Ding et al. [2023] adapt class-conditional CP for many classes / small per-class samples; Xie et al. [2025] (concurrent) study open-set + imbalanced classification with reweighting and obtain a $O(n^{-1/2})$ marginal-coverage bound *without* $\pi_{\min}$-local product. None of these address the heteroscedastic axis: in our H2H ablation (Tab. 16), single-axis Mondrian predictors face a fundamental tradeoff in the imbalanced regime — class-Mondrian preserves $\mathrm{cov}_{|Y=1}$ but loses bin-local coverage (gap 0.32–0.45), while $\hat{\sigma}$-Mondrian preserves local coverage but loses $\mathrm{cov}_{|Y=1}$ down to 0.66 at $\pi_{\min} = 0.10$. HCCP's joint ($y \times \hat{\sigma}$-bin) Mondrian is, to our knowledge, the first construction simultaneously satisfying both objectives in this regime, with T5 quantifying the price as $O(\sqrt{L_F R/(\pi_{\min} n)})$.

Heteroscedastic UQ and VEP benchmarks. Nix and Weigend [1994] originated Gaussian NLL for learned variance; Lakshminarayanan et al. [2017] introduced deep ensembles. **DEGU** [Zhou et al., 2026] pioneered ensemble distillation to a Gaussian-uncertainty student for MPRA regression and motivated conformal prediction as future work; we extend its $\hat{\sigma}$ machinery to (a) variant classification and (b) finite-sample local coverage with adaptive partition granularity. We do not claim a head-to-head defeat of DEGU: DEGU is a regression pipeline on MPRA with a Keras CNN ensemble, whereas HCCP targets binary classification on TraitGym/ProteinGym; the "DEGU-lite" ablation in App. C reimplements the ensemble-distilled $\hat{\sigma}$ at our matched GBM backbone solely to isolate the choice of variance source, not to replicate DEGU's published pipeline. **TraitGym** [Benegas et al., 2025] standardizes VEP evaluation; features include CADD [Kircher et al., 2014], Borzoi [Linder et al., 2025], GPN-MSA [Benegas et al., 2023]; AlphaGenome [Avsec et al., 2026] is a recent foundation model used in our App. C ablation. CP applications in genomic medicine [Papangelou et al., 2025, Wang et al., 2025, Adegoke et al., 2025] target drug response or polygenic risk scores in a regression / clinical-screening setting; HCCP is, to our knowledge, the first CP framework with formal class-conditional and bin-local coverage guarantees for binary variant pathogenicity.

3 Problem formulation

3.1 Setup and assumptions

Let $\mathcal{D} = \{(X_i, Y_i, C_i)\}_{i=1}^n$ with $X_i \in \mathcal{X}$, $Y_i \in \{0, 1\}$, and chromosome label $C_i \in \mathcal{C}$. We use the chrom-LOO protocol of Benegas et al. [2025]: a test chromosome $c^\star \in \mathcal{C}$ is held out and the calibration fold $\mathcal{D}^{(-c^\star)} := \{i : C_i \neq c^\star\}$ is used for both the aggregator $\hat{p}$ and the heteroscedastic head $\hat{\sigma}$. Test point $(X, Y) \sim P_{c^\star}$.

Assumption 1 (A1 chrom-group exchangeability)**.** *Variants from the same chromosome are exchangeable:* $\{(X_i, Y_i) : C_i = c\} \overset{d}{=} \{(X_{\pi(i)}, Y_{\pi(i)}) : C_i = c\}$ *for any permutation* π *within chromosome* c.

Assumption 2 (A1′ chrom-wise i.i.d.)**.** $\{(X_i, Y_i) : C_i = c\} \overset{i.i.d.}{\sim} P_c$.

Assumption 3 (A2 score stationarity across chromosomes)**.** *A2-marginal: score law is chrom-invariant. A2-class: invariant conditional on* $Y = k$*. A2-cell: invariant conditional on* $Y = k$ *and* $\hat{\sigma}$*-bin. Hierarchy: A2-cell* $\Rightarrow$ *A2-class* $\Rightarrow$ *A2-marginal.*

Empirically, the cell-level KS audit (App. B, Tab. 4) finds approximate stationarity on Complex (10.4% rejection across 125 tests, near the 5% null) and pronounced cell-level violations on Mendelian (46.2% rejection across 52 tests, max KS 0.434); T3′ is the operational certificate on Mendelian (§5.1), T3 exact suffices on Complex.

Assumption 4 (A-SL Score-Lipschitz regularity)**.** *For each class* k, $\sup_s |F_{k,\sigma_1}(s) - F_{k,\sigma_2}(s)| \leq L_F|\sigma_1 - \sigma_2|$ *for all* $\sigma_1, \sigma_2 \in [\hat{\sigma}_{\min}, \hat{\sigma}_{\max}]$.

4 is used only by T5 (§5.2) to derive the optimal bin count $K^\star$. Empirical audit of 4 via per-pair Lipschitz ratios on $\hat{\sigma}$-bin pairs (App. B, Fig. 4) gives noise-adjusted upper envelope $L_F^{(95\%)} = 13.8$ (Mendelian) and 8.1 (Complex); LCLS estimator [Huang et al., 2024] gives a tighter typical-slope estimate $\hat{L}_F = 2.97$ / 4.51.

3.2 Score and prediction set

Aggregator $\hat{p}$ (HistGradientBoosting depth 2, class-balanced) and **heteroscedastic head** $\hat{\sigma}$ (same backbone, Gaussian NLL on chrom-LOO OOF residuals $r_i = |Y_i - \hat{p}^{(-C_i)}(X_i)|$). **Nonconformity score** $s(x, y) := |y - \hat{p}(x)|/(\hat{\sigma}(x) + \epsilon)$, $\epsilon = 10^{-6}$. **Mondrian prediction set**: partition $\mathcal{D}^{(-c^\star)}$ into $2K$ cells via $(Y_i, \mathrm{bin}_K(\hat{\sigma}(X_i)))$ where bin_K uses calibration-fold $\hat{\sigma}$-quantiles. Per-cell threshold:

$$\hat{q}_{k,b} := \mathrm{Quantile}\big(\{s(X_i, Y_i) : Y_i = k, \mathrm{bin}_K(\hat{\sigma}(X_i)) = b\},\ \lceil (n_{k,b} + 1)(1 - \alpha)\rceil / n_{k,b}\big). \quad (1)$$

The Mondrian prediction set:

$$\mathcal{C}_\alpha(x) := \{k \in \{0, 1\} : |k - \hat{p}(x)| \leq \hat{q}_{k,\mathrm{bin}_K(\hat{\sigma}(x))} \cdot (\hat{\sigma}(x) + \epsilon)\}. \quad (2)$$

For cells with $n_{k,b} < \lceil 1/\alpha \rceil$ we fall back to the pooled class-conditional threshold $\hat{q}_k$ (T2).

4 Method: HCCP

Pipeline. HCCP is a three-stage post-hoc framework on top of any probabilistic classifier: (1) aggregator $\hat{p}$, (2) heteroscedastic head $\hat{\sigma}$, (3) Mondrian-($y \times \hat{\sigma}$-bin) conformal calibration. Each stage is decoupled, so HCCP accepts any $(\hat{p}, \hat{\sigma})$ pair as a drop-in. Algorithm 1 states the full pipeline; per-cell small-cell fallback to the pooled T2 threshold triggers when $n_{k,b} < \lceil 1/\alpha \rceil$.

Why the joint ($y \times \hat{\sigma}$**-bin**) **partition.** Pure class-conditional calibration (Mondrian on y alone) [Vovk et al., 2003] fixes the imbalance pathology of split CP but leaves within-class local coverage unaddressed: our ablation reaches $\mathrm{cov}_{|Y=1} = 0.896$ on Complex but the hardest $\hat{\sigma}$-decile has coverage 0.454. Stratifying by $\hat{\sigma}$-bin alone re-introduces minority-class pathology because $\hat{\sigma}$-distribution is dominated by negatives. Among Mondrian partitions over $\{y, b(x)\}$, the product partition $(y \times b)$ is the unique refinement that admits both T2 (which requires conditioning on y) and T3 (which requires conditioning on b); the two strict coarsenings each violate one of the two coverage targets.

K **selection: nested chrom-LOO.** We select $\hat{K}(c_{\mathrm{outer}})$ per outer fold via *proper nested chrom-LOO cross-validation*, ensuring $\hat{K}$ is a deterministic function of the outer calibration set alone (CP exchangeability-preserving). For each outer fold, run inner chrom-LOO and pick $\hat{K}(c_{\mathrm{outer}}) = \mathrm{argmin}_K\, \overline{G}_{\mathrm{inner}}(K)$ where the inner-fold worst-cell gap is measured on a fixed-K_{eval} sigma-bin partition (not on each candidate K itself, which would artificially favour smaller K via fewer cells). K_{eval} is set per dataset to keep ≥ 100 minority-class samples per cell: $K_{\mathrm{eval}} = 3$ on Mendelian, $K_{\mathrm{eval}} = 5$ on Complex. Modal $\hat{K}(c_{\mathrm{outer}})$ is 5 on Complex and 5–8 on Mendelian (App. C).

Algorithm 1 HCCP — Heteroscedastic Class-Conditional Conformal Prediction

Require: Data $\mathcal{D}$, test chrom $c^\star$, miscoverage α, bin count K

$\mathcal{D}_{\text{tr}} \leftarrow \{i : C_i \neq c^\star\}$; fit $\hat{p}$ on $\mathcal{D}_{\text{tr}}$ with chrom-LOO OOF predictions $\{\hat{p}^{(-C_i)}(X_i)\}$
Fit $\hat{\sigma}$ on $\mathcal{D}_{\text{tr}}$ with targets $\{|Y_i - \hat{p}^{(-C_i)}(X_i)|\}$ under Gaussian NLL
Compute $\hat{\sigma}$-bin edges via calibration-fold quantiles; calibration scores $s_i = |Y_i - \hat{p}(X_i)|/(\hat{\sigma}(X_i) + \epsilon)$
Partition $\mathcal{D}_{\text{tr}}$ by $(Y_i, \text{bin}_K(\hat{\sigma}(X_i)))$; compute $\hat{q}_{k,b}$ via Eq. (1)
for cells with $n_{k,b} < \lceil 1/\alpha \rceil$ **do**
    $\hat{q}_{k,b} \leftarrow \hat{q}_k$ ▷ T2 pooled class-conditional fallback
end for
for $x \in \mathcal{D}_{\text{te}}$ **do**
    $\mathcal{C}_\alpha(x) \leftarrow \{k \in \{0,1\} : |k - \hat{p}(x)| \leq \hat{q}_{k,b(x)} \cdot \hat{\sigma}(x)\}$
end for
return $\{(x, \mathcal{C}_\alpha(x))\}$

5 Theory

This section states HCCP's four main-text theorems. Auxiliary lemmas (T1/T2 corollaries; T1′/T2′/T4 robust counterparts via Barber et al. [2023] Thm. 2; T3-loc feature-ball extension; T3.b $\hat{\sigma}$-perturbation robustness) are stated and proved in App. A.

5.1 Bin-conditional coverage (T3 / T3′)

Theorem 1 (T3 bin-conditional coverage, exact)**.** *Under 2 and A2-cell, for each cell with $n_{k,b} \geq \lceil 1/\alpha \rceil$ and continuous F_{kb}, $1 - \alpha \leq P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k,\, b(X) = b\big) \leq 1 - \alpha + 1/(n_{k,b} + 1)$.*

Positioning vs. Barber et al. [2020]. We do not claim pointwise conditional coverage $P(Y \in \mathcal{C}_\alpha(X) \mid X = x) \geq 1 - \alpha$, ruled out in finite samples without structural assumptions. T3 conditions on the (y, b)-partition event; the $\hat{\sigma}$-bins are the coarsest practically useful strengthening of standard class-conditional CP that still carries finite-sample bounds.

Theorem 2 (T3′ bin-conditional coverage, robust)**.** *Let $\bar{\delta}_{\text{TV}}^{(k,b)}$ be the average TV distance between calibration-fold and test-fold conditional score distributions in cell (k, b). Under 2 alone (without A2-cell), for each cell with $n_{k,b} \geq \lceil 1/\alpha \rceil$, $P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k,\, b(X) = b\big) \geq (1 - \alpha) - \bar{\delta}_{\text{TV}}^{(k,b)}$.*

T3′ is the operative *certificate* on Mendelian rather than a tight prediction: empirical $\bar{\delta}_{\text{TV}}^{(k,b)} \leq 0.31$ on the worst cell yields a coverage lower bound of 0.59, which is loose by construction (uniform over all distributions satisfying 2 alone), while the observed per-chromosome coverage gap remains ≤ 0.073 throughout. T3′ supplies the only uniform finite-sample lower bound on coverage when T3's exactness assumption (A2-cell) is empirically rejected. Our per-chromosome KS audit (App. B) shows the rejection rate is approximately K-invariant across $K_{\text{eval}} \in \{2, 3, 5, 8, 10\}$ (40–46% on Mendelian), so T3′ supplies the certificate at every partition granularity.

5.2 Optimal bin count and matching lower bound (T5.1, T5.2)

This is the principal theoretical contribution. T5.1 derives the optimal K by formalising the bias-variance tradeoff and T5.2 proves a matching finite-sample lower bound *within the equi-bin Mondrian-K family* to establish that the $O(n^{-1/2})$ rate is tight on this class, dimension-free, and depends explicitly on $\pi_{\min}$ via the $\pi_{\min}^{-1/2}$ constant. The lower bound is not a distribution-free minimax bound over arbitrary CP procedures (other axes — e.g. self-calibrating CP [van der Laan, 2024] — achieve different rates on different metrics).

Under 4, the worst-cell coverage gap decomposes as

$$G(K) := \max_{k,b} |\text{cov}_{k,b} - (1 - \alpha)| \leq \underbrace{\frac{L_F R}{K}}_{\text{stationarity bias}} + \underbrace{\frac{K}{\pi_{\min} n}}_{\text{finite-sample variance}}, \tag{3}$$

where $R = \hat{\sigma}_{\max} - \hat{\sigma}_{\min}$ and $\pi_{\min} = \min(\pi, 1-\pi)$.

Theorem 3 (T5.1 oracle bin count)**.** *Under 2 and 4 with equiprobable $\hat{\sigma}$-bins, $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$ minimises Eq.* (3)*, with $G(K^\star) \le 2\sqrt{L_F R/(\pi_{\min} n)}$.*

Theorem 4 (T5.2 matching finite-sample lower bound on the equi-bin Mondrian-K class)**.** *Restrict attention to conformal predictors using equiprobable $\hat{\sigma}$-bin Mondrian partitions on the calibration fold (the class HCCP belongs to). There exists a distribution satisfying 2 and 4 such that, for any K within this class, $\min_K G(K) = \Omega\big(\sqrt{L_F R/(\pi_{\min} n)}\big)$. Hence the $O(n^{-1/2})$ rate of T5.1 is tight on this class, and the $\pi_{\min}^{-1/2}$ constant is not improvable without strengthening assumptions or moving outside the class.*

Proof sketch. For T5.1: $g'(K) = -L_F R/K^2 + 1/(\pi_{\min} n) = 0$ gives $K^\star = \sqrt{L_F R \pi_{\min} n}$. For T5.2: a linear-shift family $F_{k,\sigma}(s) = \Phi(s - L_F \sigma)$ matches both bias and variance lower bounds. Full proofs in App. A.11.

Practical $\hat{K}$ via nested chrom-LOO. Modal $\hat{K}(c_{\text{outer}}) = 5$ Complex / $\{5, 8\}$ Mendelian. Plug-in LCLS $\hat{L}_F$ [Huang et al., 2024] gives asymptotic $K^\star = 22$ (Mendelian) / 50 (Complex), above the modal nested-CV $\hat{K}$; the gap is the finite-sample fallback effect (App. C). The synthetic n-sweep (App. F) verifies the $O(n^{-1/2})$ rate: log-log slope -0.50 ± 0.05 across feature dimensions $d \in \{8, 32, 128\}$, confirming dimension-free.

5.3 Summary of guarantees

Table 1: HCCP coverage guarantees at a glance. Auxiliary lemmas (T1, T2, T1′, T2′, T3-loc, T3.b, T4) live in App. A.

Result	Assumptions	Guarantee
T3 (Thm. 1)	A1′ + A2-cell	$\text{cov}_{k,b} \in [1-\alpha, 1-\alpha+1/(n_{k,b}+1)]$
T3′ (Thm. 2)	A1′ only	$\text{cov}_{k,b} \ge (1-\alpha) - \bar{\delta}_{\text{TV}}^{(k,b)}$
T5.1 (Thm. 3)	A1′ + 4	$G(K^\star) \le 2\sqrt{L_F R/(\pi_{\min} n)}$, $K^\star = O(\sqrt{n})$
T5.2 (Thm. 4)	equi-bin Mondrian-K	$\min_K G(K) = \Omega(\sqrt{L_F R/(\pi_{\min} n)})$

6 Experiments

Setup. TraitGym Mendelian matched_9 ($n = 3{,}380$, 113 OMIM) and Complex matched_9 ($n = 11{,}400$, 29 traits); chrom-LOO with test chroms $\{17{-}22, X\}$ (TraitGym convention). Features: CADD [Kircher et al., 2014], Borzoi [Linder et al., 2025], GPN-MSA [Benegas et al., 2023]. AUPRC chrom-weighted, $\alpha = 0.10$. HCCP partition $K = \hat{K}(c_{\text{outer}})$ via nested chrom-LOO; reporting metric on fixed K_{eval} (Mendelian $K_{\text{eval}} = 3$, Complex $K_{\text{eval}} = 5$, set per dataset to keep ≥ 100 minority-class samples per cell). **Coverage and gap statistics throughout the H2H comparison are $B = 200$ chromosome-level bootstrap mean $\pm$ std for HCCP and all matched-$(\hat{p}, \hat{\sigma})$ baselines (Tab. 16, Tab. 17).** The aggregator backbone comparison in Tab. 2 reports single-seed point estimates for the non-HCCP backbones (different aggregator pipelines); HCCP-side bootstrap CIs are unchanged from Tab. 16 (full setup in App. B).

6.1 Main results

The GBM aggregator alone reaches SOTA Mendelian AUPRC 0.900, a $+14.8$ pp improvement over the published LogReg baseline [Benegas et al., 2025]. Class-conditional calibration fixes the minority-class pathology of split CP ($\text{cov}_{|Y=1}$ $0.54 \to 0.90$ on Complex) but leaves the local gap at 0.46–0.51. HCCP's joint ($y \times \hat{\sigma}$-bin) Mondrian reduces the $\hat{\sigma}$-bin gap by a further 5–25$\times$, to within one standard error of the finite-sample $1/(n_{k,b}+1)$ floor.

6.2 Head-to-head against contemporary CP

We compare HCCP (B4) against six contemporary CP variants at matched $(\hat{p}, \hat{\sigma})$: split (B1), $\hat{\sigma}$-Mondrian (B2, [Boström and Johansson, 2020]), class-Mondrian (B3, [Sadinle et al., 2019]), RLCP

Table 2: TraitGym chrom-LOO. HCCP at $K = 5$ for direct comparability with split-CP and class-conditional baselines. AUPRC chrom-weighted; $\hat{\sigma}$-bin gap = max-over-bins absolute deviation from 0.90.

Dataset	Aggregator	Calibration	AUPRC ↑	Cov|$Y{=}1$	$\hat{\sigma}$-bin gap ↓	Singleton
Mendelian	LogReg [Benegas et al., 2025]	Split CP	0.752	0.89	0.54	91%
	GBM (ours)	Class-cond. (Day 10)	**0.900**	0.90	0.46	92%
	GBM	DEGU-lite + split	0.901	0.88	0.43	85%
	GBM	**HCCP (ours)**	**0.900**	$0.894_{\pm 0.015}$	$\mathbf{0.020}_{\pm 0.012}$	$0.776_{\pm 0.040}$
Complex	LogReg	Split CP	0.332	0.88	0.43	23%
	GBM	Class-cond. (Day 10)	0.353	0.90	0.51	31%
	GBM	DEGU-lite + split	0.353	0.86	0.43	20%
	GBM	**HCCP (ours)**	0.353	$0.904_{\pm 0.004}$	$\mathbf{0.004}_{\pm 0.002}$	$0.228_{\pm 0.019}$

[Hore and Barber, 2025], weighted CP [Tibshirani et al., 2019], and SC-CP [van der Laan, 2024]. **At the recommended operating point ($K_{\text{eval}} = 3$ Mendelian, 5 Complex; per-cell minority ≥ 100), HCCP is the only single-fold partition we evaluated that holds $\text{cov}_{|Y=1} \geq 0.85$ and $\hat{\sigma}$-bin gap ≤ 0.20 on both datasets** (Tab. 16, Tab. 17; all rows $B = 200$ chrom-bootstrap mean $\pm$ std). HCCP tightens the bin-local gap by $13.7\times$ on Complex / $2.3\times$ on Mendelian over the on-axis baseline B2 $\hat{\sigma}$-Mondrian (HCCP $0.060/0.173$ vs. B2 $0.827/0.401$, bootstrap means), and $13.4\times/2.4\times$ over the off-axis B3 class-Mondrian. The two ratios coincide because B2 $\hat{\sigma}$-stratification still loses minority coverage ($\text{cov}_{|Y=1} = 0.624$ on Complex), so its $\hat{\sigma}$-bin gap is dominated by minority-cell undercoverage — the joint $(y \times \hat{\sigma})$ product, not the $\hat{\sigma}$ axis alone, is what binds. The Mendelian advantage narrows under weighted CP (HCCP 0.173 ± 0.192 vs. wCP 0.252 ± 0.153, $1.5\times$); a $K_{\text{eval}} \in \{2, 3, 5, 8, 10\}$ sweep (App. C.1) shows HCCP dominates the full range on Complex (3–$34\times$) but reverses on Mendelian at $K_{\text{eval}} \geq 5$ (per-cell minority < 70). The same calibration on the published LogReg aggregator yields essentially identical $\hat{\sigma}$-bin gap (App. C.8), decoupling the conformal-side gain from base-predictor accuracy.

Imbalance scaling validates T5's $\pi_{\min}$ dependence. Subsampling Complex to $\pi_+ \in \{0.05, 0.10, 0.20, 0.30, 0.50\}$, B1 and B2 minority-coverage degrades monotonically as $\pi_+ \to 0$ ($0.587 \to 0.810$ for B1; $0.650 \to 0.878$ for B2), while HCCP and class-Mondrian hold at the target across the full sweep (App. C, Tab. 18) — the empirical realisation of the $\pi_{\min}^{-1/2}$ factor in T5's bound.

6.3 Three-axis OOD + cross-domain stress test

We evaluate HCCP on three orthogonal distributional axes (chrom-LOO, trait-LOO, cross-dataset) and a cross-domain benchmark (ProteinGym). *Trait-LOO* reaches the finite-sample $1/(n_{k,b} + 1)$ floor on both datasets ($\hat{\sigma}$-bin gap ≤ 0.004), with one honest exception: the Skeletal/connective OMIM cluster ($n_{\text{tot}} = 470$) shows $\text{cov}_{|Y=1} = 0.745$ ($1/3$ Mendelian clusters fall below the 0.85 minority-coverage target). *Chrom-LOO* on the SOTA feature set: gap ≤ 0.077 Mendelian and 0.023 Complex. *Cross-dataset* C→M holds (gap 0.035); the reverse M→C direction is reported as a failure-mode case study (§6.4). *ProteinGym cross-domain* [Notin et al., 2023]: across 50 assay-LOO held-out proteins, HCCP attains marginal coverage 0.887, within ± 0.02 of target on $27/50$ assays and within ± 0.05 on $44/50$. The 6 outlier assays correlate with low base-predictor AUPRC (Spearman $\rho = 0.46$, $p = 0.001$), confirming coverage failures track base-predictor weakness rather than conformal failure (full diagnosis in App. E).

6.4 Failure mode: M→C cross-dataset

The reverse cross-dataset direction (Mendelian-trained → Complex-tested) jointly violates HCCP's two preconditions: (i) the base predictor collapses (AUPRC 0.205 vs. in-distribution 0.353), and (ii) calibration-fold exchangeability fails at the dataset level (ClinVar pathogenicity vs. GWAS catalog selection criteria; Tibshirani et al. [2019]). Marginal coverage drops to 0.738, within 1.4 std of the Barber 2023 Thm 2 marginal-coverage proxy bound (0.742). We do not claim T3′ binds the bin-conditional gap in this regime; this is the joint-failure boundary that delineates where HCCP's coverage machinery is operational. The forward C→M direction (in §6.3) and ProteinGym cross-domain do not violate either precondition, and HCCP holds.

6.5 Ablations

Ablations on K-sweep (T5 rate validation), $\hat{\sigma}$ source (supervised vs. DEGU-lite ensemble), $\hat{\sigma}$ target function (residual vs. log-variance), nonconformity score, class-conditional on/off, AlphaGenome [Avsec et al., 2026] feature extension, and synthetic n-sweep (verifying the $O(n^{-1/2})$ rate of Theorem 3) are reported in App. C and App. F. The synthetic sweep is the direct test of T5.1's dimension-free claim: log-log slope -0.50 ± 0.05 across $d \in \{8, 32, 128\}$ matches the $\pi_{\min}^{-1/2}$ predicted constant. The DEGU-lite ablation [Zhou et al., 2026] swaps our supervised residual $\hat{\sigma}$ for an ensemble-distilled Gaussian-uncertainty $\hat{\sigma}$ at matched GBM backbone and matched HCCP-style Mondrian calibration; this isolates the choice of variance source within HCCP and is *not* a head-to-head against DEGU's published Keras CNN MPRA pipeline. With this caveat, the supervised residual head outperforms the distilled ensemble head by $4.3\times$ on Complex worst-cell coverage, identifying where aleatoric-aware $\hat{\sigma}$ binds (polygenic complex traits) and where ensemble disagreement suffices (monogenic Mendelian).

7 Discussion

Clinical implications. HCCP prediction sets are not a replacement for expert review; they are a calibrated abstention signal. A singleton set $\mathcal{C}_\alpha(x) = \{k\}$ is a confident call; a set containing both labels or neither is an explicit abstention. On TraitGym Mendelian, HCCP produces singletons for $\approx 78\%$ of variants at $\alpha = 0.10$. Operating-point selection by miscoverage level is dataset-specific: $\alpha = 0.10$ for high-stakes rare-disease classification, $\alpha = 0.20$ for population-screening triage of complex traits where the abstention burden is otherwise prohibitive (cf. Adegoke et al. [2025] report 41% abstention at $\alpha \approx 0.20$ in a clinical screening setting); App. D reports the singleton-vs-abstention tradeoff across $\alpha \in [0.05, 0.50]$.

Assumption discussion. 1 (chrom-group exchangeability) is load-bearing. T3′ supplies the certificate when our per-chromosome KS audit detects A2-cell violations on Mendelian (App. B). 4 (score-Lipschitz) is required only for T5; T3 / T3′ are valid without it. **Limitations**: (i) the marginal-score TV proxy underestimates swap-TV by ≈ 0.036 on M→C (App. B); (ii) the oracle $K^\star$ is large-sample; the binding constraint at $n \leq 11{,}400$ is the per-cell floor $n_{k,b} \geq \lceil 1/\alpha \rceil$, with modal nested-CV $\hat{K}(c_{\text{outer}}) \in \{5, 8\}$ Mendelian and 5 Complex (App. C); (iii) DEGU CNN end-to-end port is out of scope (200 bp DNA windows lack pathogenicity-relevant signal); we ablate the $\hat{\sigma}$-mechanism only at matched GBM backbone (DEGU-lite); (iv) ProteinGym 6/50 outliers identify per-assay $\hat{\sigma}$ adaptation as future work.

Broader impact. VEP systems inform consequential clinical decisions (cascade testing, reproductive choices). HCCP provides calibrated abstention but does not guarantee clinical safety: the $\approx 78\%$ singletons on Mendelian still carry a 10% finite-sample miscoverage floor, and abstention must be paired with expert review to translate coverage into clinical utility. Population-stratified deployment requires re-calibration on ancestry-conditional folds (a fourth Mondrian axis); TraitGym is built from ClinVar and the GWAS catalog and over-represents European-ancestry individuals. The ACMG/AMP consensus [Richards et al., 2015] treats computational predictions as PP3/BP4 supporting evidence, not stand-alone classification; HCCP singletons should be integrated into multi-evidence frameworks, not deployed as diagnostic.

8 Conclusion

We introduced HCCP, a post-hoc framework for variant effect prediction that pairs a learned heteroscedastic head $\hat{\sigma}$ with Mondrian-$(y \times \hat{\sigma}\text{-bin})$ conformal calibration. Three contributions: (1) a finite-sample $O(n^{-1/2})$ rate within the equi-bin Mondrian-K class with matching lower bound and explicit $\pi_{\min}^{-1/2}$ constant (T5.1, T5.2); (2) the HCCP framework, with T3 exact bin-conditional coverage and a Barber-2023 robustness corollary T3′ that supplies a non-vacuous (loose-by-construction) worst-case certificate when cell-level exchangeability is empirically rejected; (3) empirical validation on TraitGym Mendelian / Complex with $B = 200$ chromosome-level bootstrap CIs — at the recommended operating point ($K_{\text{eval}} = 3$ Mendelian, $K_{\text{eval}} = 5$ Complex), HCCP is the only single-fold partition we evaluated, among normalized split, $\hat{\sigma}$-Mondrian, class-Mondrian, RLCP, weighted CP,

and SC-CP, that simultaneously satisfies class-conditional and bin-local coverage at $\pi_{\min} = 0.10$. The conformal-side gain is decoupled from the choice of base predictor (HCCP-on-LogReg gives an essentially identical $\hat{\sigma}$-bin gap, App. C.8), with cross-domain replication on ProteinGym and synthetic dimension-free rate verification. Natural extensions: end-to-end joint training of $(\hat{p}, \hat{\sigma})$, richer base predictors (AlphaGenome [Avsec et al., 2026]), and population-stratified Mondrian for ancestry-conditional coverage.

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A Full proofs of T1–T5

A.1 Assumptions hierarchy

We collect in one place the assumptions invoked across §5. The chain of A2 variants is:

$$\underbrace{\text{A2-cell}}_{\text{T3}} \implies \underbrace{\text{A2-class}}_{\text{T2}} \implies \underbrace{\text{A2-marginal}}_{\text{T1}},$$

where each level is obtained from the stronger one by marginalizing over a partition variable.

- **A1** (chrom-group exchangeability, 1): conditional on $C = c$, $\{(X_i, Y_i) : C_i = c\}$ are exchangeable with law P_c; different chromosomes are mutually independent.
- **A1′** (chrom-wise i.i.d., 2): the stronger version in which $\{(X_i, Y_i) : C_i = c\}$ are i.i.d. P_c.
- **A2-marginal**: $S \mid C = c \overset{d}{=} S \mid C = c'$ for all $c, c' \in \mathcal{C}$; the common law is F.
- **A2-class**: $S \mid (Y = k,\, C = c) \overset{d}{=} S \mid (Y = k,\, C = c')$ for all $c, c',\, k \in \{0, 1\}$; common law F_k.
- **A2-cell**: $S \mid (Y = k,\, b(X) = b,\, C = c) \overset{d}{=} S \mid (Y = k,\, b(X) = b,\, C = c')$ for all $c, c',\, k,\, b$; common law F_{kb}.
- **A3-loc** (local stationarity, used in T3-loc): for every (x_0, r) satisfying the T3-loc resolution condition, $s(X, k) \mid (Y = k,\, b(X) = b,\, X \in B_\ell(x_0, r))$ is distributed as F_{kb}.
- **B1** ($\hat{\sigma}$ relative error, used in T3.b): $\|\hat{\sigma} - \sigma^\star\|_\infty / \inf_x \sigma^\star(x) \le \eta$ for some $\eta \in (0, 1)$.
- **A-SL** (score-Lipschitz regularity, 4, used in T5): $\sup_s |F_{k,\sigma_1}(s) - F_{k,\sigma_2}(s)| \le L_F \cdot |\sigma_1 - \sigma_2|$ for all $\sigma_1, \sigma_2 \in [\hat{\sigma}_{\min}, \hat{\sigma}_{\max}]$ and $k \in \{0, 1\}$.

Empirical validation of A2. The within-cell KS audit (Tab. 4) ran 177 cell-level two-sample KS tests across both datasets ($K_{\text{eval}} = 5$ bins $\times$ 2 classes $\times$ test chromosomes with non-empty cells; 52 tests on Mendelian, 125 on Complex). The rejection rates at nominal $p < 0.05$ are 46.2% (Mendelian, max KS 0.434) and 10.4% (Complex, max KS 0.315). Since $d_{\text{TV}} \le d_{\text{KS}}/\sqrt{2}$ does *not* hold in general — only $d_{\text{TV}} \le d_{\text{KS}}$ for univariate F, with the tighter $\sqrt{2}$ bound under additional regularity — we use the conservative envelope $\bar{\delta}_{\text{TV}}^{(k,b)} \le d_{\text{KS}}$, giving worst-case $\bar{\delta}_{\text{TV}}^{(k,b)} \le 0.31$ on the binding Mendelian cell (chr9, chr7, chr3 pathogenic-rich at the highest-$\hat{\sigma}$ bin). A2-cell holds approximately on Complex; T3′ supplies the certificate on Mendelian.

A.2 Auxiliary lemmas

We state four standard lemmas used in the proofs below.

Lemma 1 (Rank uniformity for exchangeable sequences)**.** *Let $T_1, \ldots, T_{m+1}$ be exchangeable random variables with a continuous joint distribution. Then the rank of T_{m+1} among $(T_1, \ldots, T_{m+1})$ is uniformly distributed on $\{1, \ldots, m+1\}$.*

Proof. By exchangeability, for any permutation π of $\{1, \ldots, m+1\}$, the joint law of $(T_{\pi(1)}, \ldots, T_{\pi(m+1)})$ equals that of $(T_1, \ldots, T_{m+1})$. Continuity implies $P(T_i = T_j) = 0$ for $i \ne j$, so the ranking is almost surely unique. For each $r \in \{1, \ldots, m+1\}$, the event $\{\text{rank}(T_{m+1}) = r\}$ can be mapped to the event $\{\text{rank}(T_{\pi(m+1)}) = r\}$ by a suitable permutation that places the element currently at rank r in position $m+1$, so all rank values are equiprobable. $\square$

Lemma 2 (Exchangeable CP quantile bound)**.** *Under the conditions of Lemma 1, let* $\hat{q} = \mathrm{Quantile}(\{T_1, \ldots, T_m\}, \lceil (m+1)(1-\alpha) \rceil / m)$ *be the* $\lceil (m+1)(1-\alpha) \rceil$*-th order statistic of the first* m *values. Then*

$$P(T_{m+1} \le \hat{q}) \;=\; \frac{\lceil (m+1)(1-\alpha) \rceil}{m+1} \;\in\; \left[1-\alpha,\; 1-\alpha+\frac{1}{m+1}\right].$$

Proof. By Lemma 1, T_{m+1} is at rank r with probability $1/(m+1)$ for each r. The event $\{T_{m+1} \le \hat{q}\}$ is exactly $\{\mathrm{rank}(T_{m+1}) \le \lceil (m+1)(1-\alpha) \rceil\}$ (since $\hat{q}$ is the $\lceil (m+1)(1-\alpha) \rceil$-th order statistic of $(T_1, \ldots, T_m)$, which is also the $\lceil (m+1)(1-\alpha) \rceil$-th order statistic of the augmented set when T_{m+1} has rank at most that value). Thus

$$P(T_{m+1} \le \hat{q}) = \frac{\lceil (m+1)(1-\alpha) \rceil}{m+1}.$$

Since $\lceil (m+1)(1-\alpha) \rceil \in [(m+1)(1-\alpha),\, (m+1)(1-\alpha)+1)$, dividing by $m+1$ gives the stated bounds. □

Lemma 3 (Monotone score-rescaling invariance)**.** *Let* $g : \mathbb{R}_+ \to \mathbb{R}_+$ *be a strictly increasing function. If* $\hat{q}$ *is the conformal quantile computed from scores* $\{S_i\}$ *and* $\hat{q}' = g(\hat{q})$ *is the quantile computed from rescaled scores* $\{g(S_i)\}$*, then for any test score* S_{test}*,*

$$S_{\text{test}} \le \hat{q} \quad \Longleftrightarrow \quad g(S_{\text{test}}) \le \hat{q}'.$$

Proof. The quantile index $\lceil (m+1)(1-\alpha) \rceil$ is the same for both sets. Since g is strictly increasing, it preserves the ordering of scores. Thus the j-th order statistic of $\{g(S_i)\}$ equals g applied to the j-th order statistic of $\{S_i\}$, giving $\hat{q}' = g(\hat{q})$. The equivalence follows from g being order-preserving. □

Lemma 4 (Bin-boundary mass bound)**.** *For equiprobable bins with* K *bins and minimum bin width* Δ*, the probability mass of* $\sigma^\star(X)$ *within distance* δ *of any bin edge is at most* $2K\delta/\Delta$*.*

Proof. Each bin contains mass $1/K$ and has width $\ge \Delta$, so the density of $\sigma^\star(X)$ is at most $1/(K\Delta) \cdot K = 1/\Delta$ on average. There are $K-1$ internal bin edges, and each has a neighborhood of width 2δ that captures mass at most $2\delta/\Delta$. Summing over all edges gives $2(K-1)\delta/\Delta \le 2K\delta/\Delta$. □

A.3 Lemma 5: T1 marginal coverage

Lemma 5 (T1 marginal coverage — standard split-CP at the marginal partition level)**.** *Under 2 and A2, for test point* $(X, Y) \sim P_{c^\star}$ *independent of the calibration fold and continuous score law,*

$$1-\alpha \;\le\; P\big(Y \in \mathcal{C}_\alpha(X)\big) \;\le\; 1-\alpha+\frac{1}{n_{\min}+1}, \qquad n_{\min} = \min_k n_k.$$

Proof. We decompose the marginal coverage via the true label.

Step 1 (Reduction to per-class coverage). By the law of total probability:

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\big) = \sum_{k \in \{0,1\}} P(Y_{\text{test}} = k) \cdot P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k\big).$$

Since $Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})$ reduces to $s(X_{\text{test}}, Y_{\text{test}}) \le \hat{q}_{Y_{\text{test}}}$ by definition of $\mathcal{C}_\alpha$, we have

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k\big) = P\big(s(X_{\text{test}}, k) \le \hat{q}_k \mid Y_{\text{test}} = k\big).$$

Step 2 (Joint exchangeability inside class k*).* Fix k. Under 2, for each calibration chromosome $c \ne c^\star$, $(X_i, Y_i)_{C_i = c}$ are i.i.d. P_c. The score $S_i = s(X_i, Y_i)$ is a deterministic function of (X_i, Y_i) via the fixed $(\hat{p}, \hat{\sigma})$ trained on $\mathcal{D}^{(-c^\star)}$. Restricting to $i \in \mathcal{I}_k^{(-c^\star)}$ (i.e., $Y_i = k$) yields i.i.d. samples from a chrom-specific conditional law F_k^c. By A2-class (implied by A2-marginal when class prevalence $P(Y = k \mid C)$ is chrom-independent, or invoked directly), $F_k^c = F_k$ for all c. Across chromosomes, samples are mutually independent (A1). Hence the full calibration set $\{S_i\}_{i \in \mathcal{I}_k^{(-c^\star)}}$ is i.i.d. F_k.

Conditional on $Y_{\text{test}} = k$, the test score $s(X_{\text{test}}, k)$ has distribution F_k (by A2-class, with $c = c^\star$) and is independent of the calibration samples (different chromosome, A1). Thus the augmented sequence $(S_1^{(k)}, \ldots, S_{n_k}^{(k)}, s(X_{\text{test}}, k))$ is i.i.d. F_k, hence exchangeable.

Step 3 (Standard exchangeable CP bound). By Lemma 2 applied with $m = n_k$ and $T_{m+1} = s(X_{\text{test}}, k)$:

$$P\big(s(X_{\text{test}}, k) \le \hat{q}_k \mid Y_{\text{test}} = k\big) = \frac{\lceil (n_k+1)(1-\alpha) \rceil}{n_k+1} \in \left[1-\alpha,\ 1-\alpha+\frac{1}{n_k+1}\right].$$

Step 4 (Assemble). Substituting into Step 1:

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\big) = \sum_k P(Y_{\text{test}} = k) \cdot P\big(s(X_{\text{test}}, k) \le \hat{q}_k \mid Y_{\text{test}} = k\big).$$

Each term lies in $[1-\alpha,\ 1-\alpha+1/(n_k+1)]$. The convex combination inherits the range: the lower bound gives $\ge 1-\alpha$; the upper bound gives $\le 1-\alpha+1/(n_{\min}+1)$ with $n_{\min} = \min_k n_k$. □

A.4 Lemma 6: T2 class-conditional coverage

Lemma 6 (T2 class-conditional coverage — standard split-CP at the class partition level)**.** *Under the conditions of Lemma 5 with A2 at class level and $n_k \ge \lceil 1/\alpha \rceil$ for each $k \in \{0,1\}$,*

$$1-\alpha \ \le\ P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k\big) \ \le\ 1-\alpha+\frac{1}{n_k+1}.$$

Proof. Fix $k \in \{0,1\}$ with $n_k \ge \lceil 1/\alpha \rceil$.

Step 1 (Exchangeability). Under 2 and A2-class, conditioning on $Y_i = k$ makes $(S_1^{(k)}, \ldots, S_{n_k}^{(k)})$ i.i.d. F_k by the same argument as T1, Step 2. Conditional on $Y_{\text{test}} = k$, $s(X_{\text{test}}, k)$ is independent of calibration and distributed as F_k. Thus the joint $(S_1^{(k)}, \ldots, S_{n_k}^{(k)}, s(X_{\text{test}}, k))$ is i.i.d., hence exchangeable.

Step 2 (Quantile bound). The prediction-set inclusion event $\{Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\}$ under $Y_{\text{test}} = k$ is $\{s(X_{\text{test}}, k) \le \hat{q}_k\}$. By Lemma 2 with $m = n_k$:

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k\big) = \frac{\lceil (n_k+1)(1-\alpha) \rceil}{n_k+1} \in \left[1-\alpha,\ 1-\alpha+\frac{1}{n_k+1}\right].$$

The sample-size requirement $n_k \ge \lceil 1/\alpha \rceil$ ensures $\lceil (n_k+1)(1-\alpha) \rceil \le n_k$, so $\hat{q}_k < +\infty$ and the prediction set is non-trivial. □

Remark. T1 is the law-of-total-probability average of the two T2 per-class bounds, so T2 is strictly stronger. The sample-size requirement is comfortably met on TraitGym: at $\alpha = 0.10$ we need $n_k \ge 10$; the smallest chrom-LOO calibration has $n_1 \ge 30$ (Mendelian) and $n_1 \ge 100$ (Complex).

A.5 Robust fallback: T1′ and T2′

When A2 fails (e.g., a test chromosome has systematically different variant characteristics), we invoke the following restatement of Barber et al. [2023] Theorem 2 in our notation.

Barber 2023 Theorem 2 (restated). Let $Z = (Z_1, \ldots, Z_n, Z_{n+1})$ be the full data ordering with Z_{n+1} as the test point, $R(Z) = (S_1, \ldots, S_{n+1})$ the residual vector, and $R(Z^{(i)})$ the sequence with Z_i and Z_{n+1} swapped. For prespecified weights $w_i \in [0,1]$ with $\tilde{w}_i = w_i/(w_1 + \cdots + w_n + 1)$,

$$P\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \ \ge\ 1-\alpha-\sum_{i=1}^{n} \tilde{w}_i \cdot d_{\text{TV}}\big(R(Z),\ R(Z^{(i)})\big).$$

No factor of 2.

Proof of T1′ and T2′. Use uniform weights $w_i = 1$, so $\tilde{w}_i = 1/(n+1)$. The Barber 2023 bound becomes

$$P(\text{coverage}) \;\geq\; 1 - \alpha - \frac{1}{n+1} \sum_i d_{\mathrm{TV}}\big(R(Z),\, R(Z^{(i)})\big).$$

Under 1 alone (without A2), $R(Z^{(i)})$ differs from $R(Z)$ in the position of one residual: $S_i \leftrightarrow S_{\text{test}}$. If S_i is from chromosome c (so $S_i \sim F_c$) and S_{test} from $c^\star$ (so $S_{\text{test}} \sim F_{c^\star}$), the swap changes the marginal law of exactly two entries of the residual vector. Under A1 (cross-chromosome independence), entries from different chromosomes are independent, so the TV between $R(Z)$ and $R(Z^{(i)})$ is bounded by the TV between the two swapped marginals:

$$d_{\mathrm{TV}}\big(R(Z),\, R(Z^{(i)})\big) \;\leq\; d_{\mathrm{TV}}(F_c,\, F_{c^\star}).$$

Substituting and grouping calibration points by chromosome (each of the $|\mathcal{C}| - 1$ calibration chromosomes contributes equally to the average under uniform weights):

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\big) \;\geq\; 1 - \alpha - \bar{\delta}_{\mathrm{TV}}, \qquad \bar{\delta}_{\mathrm{TV}} = \frac{1}{|\mathcal{C}| - 1} \sum_{c \neq c^\star} d_{\mathrm{TV}}(F_c,\, F_{c^\star}).$$

This is T1′.

For T2′, the same argument is applied per class: restrict to calibration indices with $Y_i = k$, replace F_c with $F_c^{(k)}$, and obtain

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k\big) \;\geq\; 1 - \alpha - \bar{\delta}_{\mathrm{TV}}^{(k)}, \qquad \bar{\delta}_{\mathrm{TV}}^{(k)} = \frac{1}{|\mathcal{C}| - 1} \sum_{c \neq c^\star} d_{\mathrm{TV}}\big(F_c^{(k)},\, F_{c^\star}^{(k)}\big).$$

Both bounds carry no factor of 2 (weighted-swap Barber 2023, not jackknife+). □

Empirical TV estimates. The cell-level KS audit (App. B.4, 177 tests across both datasets) gives worst-cell $d_{\mathrm{KS}} = 0.434$ on Mendelian (chr9 / chr7 / chr3 highest-$\hat{\sigma}$ pathogenic-rich cell) and 0.315 on Complex. Using $\bar{\delta}_{\mathrm{TV}}^{(k,b)} \leq d_{\mathrm{KS}}$ as the conservative envelope, the cell-level coverage gap is bounded by ≤ 0.31 on Mendelian and ≤ 0.05 on Complex; observed per-chromosome coverage gaps remain ≤ 0.073 throughout, well below the certificate.

A.6 Proof of Theorem 1 (bin-conditional coverage, exact)

Proof. The argument is the class-conditional proof of Lemma 6 applied inside a single Mondrian cell (k, b), with A2-cell replacing A2-class. Fix a cell (k, b) with $n_{kb} \geq \lceil 1/\alpha \rceil$.

Step 1 (Cell restriction). The event "$Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})$ given $Y_{\text{test}} = k$" is exactly $\{s(X_{\text{test}}, k) \leq \hat{q}_{k,b(X_{\text{test}})}\}$ by the definition of $\mathcal{C}_\alpha$. Conditioning further on $b(X_{\text{test}}) = b$ fixes the threshold to $\hat{q}_{kb}$.

Step 2 (I.i.d. F_{kb} via A1′ + A2-cell). Under 2, for each calibration chromosome $c \neq c^\star$, $(X_i, Y_i)_{C_i = c}$ are i.i.d. P_c; the score $S_i = s(X_i, Y_i)$ is a deterministic function of (X_i, Y_i) via the fixed $(\hat{p}, \hat{\sigma})$. Restricting to $i \in \mathcal{I}_{kb}^{(-c^\star)}$ (i.e., $Y_i = k$ and $b(X_i) = b$) yields i.i.d. samples from a chromosome-specific conditional law, which by A2-cell equals F_{kb} for every calibration chromosome. Across chromosomes, samples are mutually independent (1). Hence the full calibration set $\{S_i\}_{i \in \mathcal{I}_{kb}^{(-c^\star)}}$ is i.i.d. F_{kb}.

Step 3 (Test-score independence and exchangeability). Conditional on $Y_{\text{test}} = k$ and $b(X_{\text{test}}) = b$, the test score $s(X_{\text{test}}, k)$ has distribution F_{kb} (by A2-cell, with $c = c^\star$) and is independent of the calibration samples (different chromosome, 1). Thus $(S_1^{(kb)}, \ldots, S_{n_{kb}}^{(kb)}, s(X_{\text{test}}, k))$ is i.i.d. F_{kb}, in particular exchangeable.

Step 4 (CP quantile bound). By Lemma 2 with $m = n_{kb}$:

$$P\big(s(X_{\text{test}}, k) \leq \hat{q}_{kb}\big) = \frac{\lceil (n_{kb} + 1)(1 - \alpha) \rceil}{n_{kb} + 1} \in \left[1 - \alpha,\; 1 - \alpha + \frac{1}{n_{kb} + 1}\right].$$

Combining Steps 1 and 4 yields

$$1 - \alpha \;\leq\; P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k,\, b(X_{\text{test}}) = b\big) \;\leq\; 1 - \alpha + \frac{1}{n_{kb} + 1}. \qquad \square$$

Local coverage gap bound. For each cell (k, b),

$$\big|P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k,\, b(X) = b\big) - (1-\alpha)\big| \;\le\; \frac{1}{n_{kb}+1}.$$

A.7 Lemma 7: T3′ bin-conditional coverage, robust

Lemma 7 (T3′ bin-conditional coverage, robust)**.** *Under 1 alone (no A2 at any level), for each cell (k, b) with $n_{k,b} \ge \lceil 1/\alpha \rceil$,*

$$P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k,\, b(X) = b\big) \;\ge\; 1-\alpha-\bar{\delta}_{\mathrm{TV}}^{(k,b)}, \qquad \bar{\delta}_{\mathrm{TV}}^{(k,b)} = \frac{1}{|\mathcal{C}|-1}\sum_{c\neq c^\star} d_{\mathrm{TV}}\big(F_c^{(k,b)},\, F_{c^\star}^{(k,b)}\big).$$

Proof. Apply Barber et al. [2023] Theorem 2 with calibration weights

$$w_i = \mathbb{1}\{i \in \mathcal{I}_{kb}^{(-c^\star)}\}, \qquad \tilde{w}_i = \frac{w_i}{w_1 + \cdots + w_n + 1}.$$

So $\tilde{w}_i = 1/(n_{kb}+1)$ if $i \in \mathcal{I}_{kb}^{(-c^\star)}$ and 0 otherwise. Barber Theorem 2 gives

$$P(\text{coverage in cell } (k,b)) \;\ge\; 1-\alpha - \sum_{i \in \mathcal{I}_{kb}^{(-c^\star)}} \tilde{w}_i \cdot d_{\mathrm{TV}}\big(R(Z),\, R(Z^{(i)})\big).$$

If i is from chromosome c (so $Z_i \sim F_c^{(k,b)}$) and Z_{test} is from $c^\star$ (so $Z_{\mathrm{test}} \sim F_{c^\star}^{(k,b)}$), the swap $Z_i \leftrightarrow Z_{\mathrm{test}}$ changes the marginal law of exactly two entries of the residual vector. Under 1 (cross-chromosome independence), entries from different chromosomes are independent, so the TV between $R(Z)$ and $R(Z^{(i)})$ is bounded by the TV between the two swapped marginals:

$$d_{\mathrm{TV}}\big(R(Z),\, R(Z^{(i)})\big) \;\le\; d_{\mathrm{TV}}\big(F_c^{(k,b)},\, F_{c^\star}^{(k,b)}\big).$$

Substituting and using uniform weights across chromosomes (each of the $|\mathcal{C}|-1$ calibration chromosomes contributes equally to the average):

$$P\big(Y_{\mathrm{test}} \in \mathcal{C}_\alpha(X_{\mathrm{test}}) \mid Y_{\mathrm{test}} = k,\, b(X_{\mathrm{test}}) = b\big) \;\ge\; 1-\alpha-\bar{\delta}_{\mathrm{TV}}^{(k,b)},$$

where

$$\bar{\delta}_{\mathrm{TV}}^{(k,b)} = \frac{1}{|\mathcal{C}|-1}\sum_{c\neq c^\star} d_{\mathrm{TV}}\big(F_c^{(k,b)},\, F_{c^\star}^{(k,b)}\big).$$

The bound has no factor of 2. □

Relation to T3. Setting $\bar{\delta}_{\mathrm{TV}}^{(k,b)} = 0$ recovers T3's lower bound exactly. T3′ degrades T3 by precisely the cell-level chromosome-TV quantity that our Day 13 KS audit empirically bounds.

A.8 Lemma 8: T3-loc feature-ball coverage

Lemma 8 (T3-loc feature-ball coverage)**.** *Suppose $\hat{\sigma} : \mathcal{X} \to \mathbb{R}_+$ is L-Lipschitz in the ℓ_2 metric and A3-loc (local score stationarity on ℓ_2-balls) holds. Let $\Delta = \min_b(e_b - e_{b-1})$ be the minimum $\hat{\sigma}$-bin width. For any $x_0 \in \mathcal{X}$, any radius r with $Lr < \Delta/2$, and any class k,*

$$\big|P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0, r),\, Y = k\big) - (1-\alpha)\big| \;\le\; \frac{1}{n_{\min}(x_0, r)+1},$$

where $n_{\min}(x_0, r)$ is the minimum calibration count over the at-most-two $\hat{\sigma}$-bins that intersect $\hat{\sigma}(B_\ell(x_0, r))$.

Proof. Assume $\hat{\sigma} : \mathcal{X} \to \mathbb{R}_+$ is L-Lipschitz and A3-loc holds. Fix $x_0 \in \mathcal{X}$, $r > 0$ with $Lr < \Delta/2$, and class k.

Step 1 (Lipschitz bound on $\hat{\sigma}$-range). For $x \in B_\ell(x_0, r)$:

$$|\hat{\sigma}(x) - \hat{\sigma}(x_0)| \;\le\; L\|x - x_0\| \;\le\; Lr.$$

So $\hat{\sigma}\big(B_\ell(x_0, r)\big) \subseteq [\hat{\sigma}(x_0) - Lr,\, \hat{\sigma}(x_0) + Lr]$.

Step 2 (Pigeonhole — at most two bins). Since $Lr < \Delta/2$, this $\hat{\sigma}$-interval has width $2Lr < \Delta$ and therefore intersects at most two adjacent bins $B_{b_\star}, B_{b_\star+1}$. (An interval of width less than the minimum bin width can straddle at most one bin boundary.)

Step 3 (Convex decomposition). Decompose the conditional coverage:

$$P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0, r),\, Y = k\big) = \sum_{b \in \{b_\star, b_\star+1\}} w_b(x_0, r) \cdot P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0, r),\, Y = k,\, b(X) = b\big),$$

where $w_b(x_0, r) := P(b(X) = b \mid X \in B_\ell(x_0, r),\, Y = k)$ are non-negative and sum to 1.

Step 4 (Within-cell restriction preserves F_{kb}). Inside each cell (k, b), the conformal threshold $\hat{q}_{kb}$ depends only on calibration and is not affected by restricting X to $B_\ell(x_0, r)$. Under A3-loc, $s(X, k) \mid (Y = k,\, b(X) = b,\, X \in B_\ell(x_0, r))$ has the cell score law F_{kb}. The T3 argument (App. A.6, Steps 2–4) then applies verbatim inside each cell restricted to the ball:

$$P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0, r),\, Y = k,\, b(X) = b\big) \in \left[1 - \alpha,\; 1 - \alpha + \frac{1}{n_{kb} + 1}\right].$$

Step 5 (Convex-hull sandwich). The convex combination over the (at most two) bins $\{b_\star, b_\star + 1\}$ lies in

$$\left[1 - \alpha,\; 1 - \alpha + \frac{1}{n_{\min}(x_0, r) + 1}\right],$$

where $n_{\min}(x_0, r) = \min_{b \in \{b_\star, b_\star+1\}} n_{kb}$. □

Honest limitations. A3-loc is stronger than A2-cell — it requires local stationarity at the scale r, which we cannot fully verify at high feature-space dimension with our $n \leq 11{,}400$. We state T3-loc conditionally on A3-loc. The Day 13 cell-level KS audit is the closest finite-sample analog. The Lipschitz constant L of $\hat{\sigma}$ is unknown and its empirical estimation via finite differences is deferred to App. B. Finally, Barber et al. [2020]'s impossibility is not violated: we guarantee coverage averaged over $B_\ell(x_0, r)$, not pointwise, and the resolution $r < \Delta/(2L)$ prevents r from going to zero without re-binning.

A.9 Lemma 9: T3.b $\hat{\sigma}$ perturbation

Lemma 9 (T3.b $\hat{\sigma}$ perturbation)**.** *Let $\sigma^\star$ be an oracle variance function and assume B1 ($\|\hat{\sigma} - \sigma^\star\|_\infty / \inf_x \sigma^\star(x) \leq \eta$). Under A1′ + A2-cell (w.r.t. the fixed bin edges $e_0 < \cdots < e_K$),*

$$\left| P\big(Y \in \mathcal{C}_\alpha^{(\hat{\sigma})}(X) \mid Y = k\big) - P\big(Y \in \mathcal{C}_\alpha^{(\sigma^\star)}(X) \mid Y = k\big) \right| \;\leq\; \frac{2\eta\,\bar{\sigma}}{\Delta} + \frac{1}{n_{\min} + 1},$$

where $\bar{\sigma} = \sup_x \sigma^\star(x)$ and $n_{\min}$ is the minimum cell size.

Proof. Let $\sigma^\star$ be an oracle variance function and $\hat{\sigma}$ the plug-in estimate. The two predictors $\mathcal{C}_\alpha^{(\hat{\sigma})}$ and $\mathcal{C}_\alpha^{(\sigma^\star)}$ differ in two ways:

(a) Bin reassignment. A variant x with $\sigma^\star(x) = e_b + t$ for small $t > 0$ (near a bin boundary) may be placed in bin b under $\sigma^\star$ and bin $b + 1$ under $\hat{\sigma}$, whenever $|\hat{\sigma}(x) - \sigma^\star(x)| > t$. Under B1, $|\hat{\sigma}(x) - \sigma^\star(x)| \leq \eta \cdot \sigma^\star(x) \leq \eta\bar{\sigma}$ where $\bar{\sigma} = \sup_x \sigma^\star(x)$. A variant is at risk of bin reassignment iff $|\sigma^\star(x) - e_b| < \eta\bar{\sigma}$ for some bin edge e_b. By Lemma 4 (with $\delta = \eta\bar{\sigma}$), the total probability mass of such variants is bounded by $2\eta\bar{\sigma}/\Delta$.

(b) Score rescaling. Within the same cell, the scores differ:

$$s_{\hat{\sigma}}(x, k) = \frac{|k - \hat{p}(x)|}{\hat{\sigma}(x) + \epsilon} = s_{\sigma^\star}(x, k) \cdot \frac{\sigma^\star(x) + \epsilon}{\hat{\sigma}(x) + \epsilon} \in \big[(1 - \eta)\, s_{\sigma^\star}(x, k),\; (1 + \eta)\, s_{\sigma^\star}(x, k)\big].$$

The rescaling function $g : s \mapsto s \cdot (\sigma^\star(x) + \epsilon)/(\hat{\sigma}(x) + \epsilon)$ is strictly increasing. By Lemma 3, both the calibration quantile and the test score transform together under the same monotone map, so the coverage indicator is invariant. Contribution (b) is exactly zero.

(c) Combining. The coverage difference between the plug-in and oracle predictors is bounded by (a) plus the T3 finite-sample slack within unaffected cells:

$$\Big|P\big(Y \in \mathcal{C}_\alpha^{(\hat\sigma)}(X) \mid Y = k\big) - P\big(Y \in \mathcal{C}_\alpha^{(\sigma^\star)}(X) \mid Y = k\big)\Big| \le \frac{2\eta\,\bar\sigma}{\Delta} + \frac{1}{n_{\min}+1}. \qquad \square$$

Graceful degradation. As $\hat\sigma \to \sigma^\star$ uniformly ($\eta \to 0$), the T3.b bound collapses to the T3 finite-sample slack $1/(n_{\min}+1)$. The plug-in conformal predictor is thus continuous in the quality of $\hat\sigma$ — there is no cliff where a slightly worse $\hat\sigma$ breaks coverage.

A.10 Lemma 10: T4 chrom-shift bound (Barber 2023 Thm 2)

Lemma 10 (T4 chrom-shift bound, Barber et al. [2023] Thm 2)**.** *For any prespecified weights $w_i \in [0,1]$ and $\tilde w_i = w_i/(w_1 + \cdots + w_n + 1)$,*

$$P\big(Y \in \mathcal{C}_\alpha(X)\big) \;\ge\; 1 - \alpha - \sum_{i=1}^{n} \tilde w_i \cdot d_{\mathrm{TV}}\big(R(Z),\ R(Z^{(i)})\big),$$

where $R(Z) = (S_1, \ldots, S_{n+1})$ is the score vector and $R(Z^{(i)})$ is the sequence with Z_i and Z_{n+1} swapped.

Proof. This is a direct restatement of Barber et al. [2023] Theorem 2 in our notation.

Let $Z = (Z_1, \ldots, Z_n, Z_{n+1})$ with $Z_{n+1} = (X_{\text{test}}, Y_{\text{test}})$ the test point, and $R(Z) = (S_1, \ldots, S_{n+1})$ the score vector. For any prespecified weights $w_i \in [0,1]$ with $\tilde w_i = w_i/(w_1 + \cdots + w_n + 1)$ and conformal predictor $\hat C_n$ built from a weighted quantile of $\{S_i\}_{i=1}^n$, Barber et al. [2023] show:

$$P\big(Y_{n+1} \in \hat C_n(X_{n+1})\big) \;\ge\; 1 - \alpha - \sum_{i=1}^{n} \tilde w_i \cdot d_{\mathrm{TV}}\big(R(Z),\ R(Z^{(i)})\big),$$

where $R(Z^{(i)})$ is the score vector with Z_i and Z_{n+1} swapped.

Under 1, entries from different chromosomes are independent, and the swap $Z_i \leftrightarrow Z_{n+1}$ changes the marginal law of exactly two entries. If Z_i comes from chromosome c and Z_{n+1} from $c^\star$:

$$d_{\mathrm{TV}}\big(R(Z),\ R(Z^{(i)})\big) \;\le\; d_{\mathrm{TV}}\big(F_c,\ F_{c^\star}\big).$$

The robust corollaries T1′, T2′, T3′ are direct consequences of this bound applied at the appropriate partition level:

- T1′: uniform weights, marginal scores, yields $\bar\delta_{\mathrm{TV}}$ (App. A.5);
- T2′: restrict to class-k calibration indices, class-conditional scores, yields $\bar\delta_{\mathrm{TV}}^{(k)}$;
- T3′: restrict to cell-(k,b) calibration indices, cell-conditional scores, yields $\bar\delta_{\mathrm{TV}}^{(k,b)}$ (App. A.7).

The bound carries no factor of 2. This is the weighted-swap construction of Barber et al. [2023], not the jackknife+ bound from an earlier Barber et al. [2021] paper. $\square$

A.11 Proof of Theorem 3 (T5.1 oracle bin count)

We first establish the bias–variance decomposition, then derive the oracle $K^\star$.

Proof of the gap decomposition (Eq. (3)*).* Under 2 and 4:

Step 1 (Bias bound). Within bin b, all $\hat\sigma$ values lie in $(e_{b-1}, e_b]$ with width Δ_b. The cell score CDF F_{kb} is a mixture:

$$F_{kb}(s) = \int_{e_{b-1}}^{e_b} F_{k,\sigma}(s)\, dG_b(\sigma),$$

where G_b is the distribution of $\hat{\sigma}$ within bin b. By 4, for any reference value $\sigma_0 \in (e_{b-1}, e_b]$:

$$\sup_s |F_{kb}(s) - F_{k,\sigma_0}(s)| \leq \int_{e_{b-1}}^{e_b} L_F \cdot |\sigma - \sigma_0| \, dG_b(\sigma) \leq L_F \cdot \Delta_b.$$

Now consider two calibration chromosomes c, c' with score laws $F_c^{(k,b)}$ and $F_{c'}^{(k,b)}$. Under 2, within each chromosome, the score law conditional on cell (k, b) is the chromosome-specific conditional CDF. If 4 holds uniformly across chromosomes (i.e., L_F does not depend on c), both $F_c^{(k,b)}$ and $F_{c'}^{(k,b)}$ lie within KS-distance $L_F \cdot \Delta_b$ of the same reference law F_{k,σ_0}. By T3′ (App. A.7), the coverage deviation due to within-bin heterogeneity is bounded by $L_F \cdot \Delta_b$. This is the stationarity bias $\beta_{kb} = L_F \cdot \Delta_b$.

Step 2 (Variance bound). Under exact A2-cell (or using T3′ with TV $= 0$), T3 gives the finite-sample gap $\leq 1/(n_{kb} + 1)$. This is the variance term $\nu_{kb} = 1/(n_{kb} + 1)$.

Step 3 (Combining). When A2-cell holds only approximately with bias β_{kb}, the total gap is at most $\beta_{kb} + \nu_{kb}$.

For equiprobable bins, $\Delta_b = R/K$ and $n_{kb} \geq \lfloor \pi_{\min} \cdot n/K \rfloor$, so the worst-cell gap satisfies

$$G(K) \leq \frac{L_F \cdot R}{K} + \frac{K}{\pi_{\min} \cdot n}.$$

(The floor is absorbed into the leading $O(n^{-1/2})$ term for the oracle.) □

Proof of Theorem 3 (T5.1). Treat K as a continuous variable. The gap function $g(K) = L_F R/K + K/(\pi_{\min} n)$ is convex, so setting $g'(K) = 0$:

$$-\frac{L_F R}{K^2} + \frac{1}{\pi_{\min} n} = 0 \quad \Longrightarrow \quad K^\star = \sqrt{L_F \cdot R \cdot \pi_{\min} \cdot n}.$$

Substituting back:

$$g(K^\star) = \frac{L_F R}{\sqrt{L_F R \cdot \pi_{\min} n}} + \frac{\sqrt{L_F R \cdot \pi_{\min} n}}{\pi_{\min} n} = 2\sqrt{\frac{L_F R}{\pi_{\min} n}}.$$

The floor $\lfloor K^\star \rfloor$ introduces an additive $O(1/(\pi_{\min} n))$ term (the gap between $K^\star$ and $\lfloor K^\star \rfloor$ shifts the variance term by at most $1/(\pi_{\min} n)$), which is dominated by the $O(n^{-1/2})$ rate. □

A.12 Proof of Theorem 4 (T5.2 matching lower bound)

Proof. We construct a distribution achieving both lower bounds simultaneously.

Bias lower bound. Consider the family $F_{k,\sigma}(s) = \Phi(s - L_F \sigma)$, where Φ is the standard normal CDF. This satisfies 4 with Lipschitz constant L_F: for any σ_1, σ_2,

$$\sup_s |F_{k,\sigma_1}(s) - F_{k,\sigma_2}(s)| = \sup_s |\Phi(s - L_F \sigma_1) - \Phi(s - L_F \sigma_2)| \leq L_F |\sigma_1 - \sigma_2|,$$

where the inequality uses $\|\Phi'\|_\infty = \phi(0) = 1/\sqrt{2\pi} < 1$.

In any bin of width $\Delta_b = R/K$, the mixture CDF F_{kb} averages over shifted CDFs. Because the shift is monotone in σ, the CDF at the lower boundary $F_{k,e_{b-1}}$ and upper boundary F_{k,e_b} straddle the mixture:

$$\sup_s |F_{kb}(s) - F_{k,e_{b-1}}(s)| \;\geq\; \frac{L_F \Delta_b}{2} = \frac{L_F R}{2K}.$$

(The factor $1/2$ comes from the average shift relative to one endpoint being at least half the bin width.)

Variance lower bound. For any cell with n_{kb} calibration points, the conformal quantile's rank-based coverage is exactly $\lceil (n_{kb} + 1)(1 - \alpha) \rceil / (n_{kb} + 1)$, which lies in $[1 - \alpha,\, 1 - \alpha + 1/(n_{kb} + 1)]$. When F_{kb} is continuous and the test point's rank is uniformly distributed, the gap $1/(n_{kb} + 1)$ is attainable (up to ceiling effects). For the worst cell, $n_{kb} \leq \lceil \pi_{\min} n/K \rceil$, so the gap is at least $1/(2\lceil \pi_{\min} n/K \rceil + 1)$.

Combined lower bound. For any K:

$$G(K) \;\geq\; \max\left(\frac{L_F R}{2K},\; \frac{1}{2\lceil \pi_{\min} n/K \rceil + 1}\right).$$

Optimizing over K: the first term is $\Theta(1/K)$ and the second is $\Theta(K/n)$. They balance at $K = \Theta(\sqrt{n})$, giving

$$\min_K G(K) = \Omega\left(\sqrt{\frac{L_F R}{\pi_{\min} n}}\right),$$

matching the $O(n^{-1/2})$ upper bound of T5.1 up to constants. □

Interpretation. The $O(n^{-1/2})$ rate is dimension-free because T5 conditions on the 1-D summary $\hat{\sigma}(x)$, not on the full feature vector. By contrast, RLCP [Hore and Barber, 2025] achieves local coverage at rate $O(n^{-2/(d+2)})$, which is impractical at $d = 7{,}731$. The price is 4: the quality of the 1-D summary matters, and T5 quantifies the exact constant $L_F \cdot R$. In the homoscedastic limit $L_F = 0$, the oracle $K^\star = 0$ and no binning is needed — T1/T2 suffice.

B Data and training details

B.1 TraitGym benchmark

We use TraitGym [Benegas et al., 2025], a standardized evaluation suite for variant effect prediction. Two matched datasets:

- **Mendelian (matched_9)**: 338 ClinVar pathogenic variants + 3,042 matched benign controls = 3,380 total across 113 OMIM phenotypes. Matching on consequence type, MAF, and TSS distance (9 strata).
- **Complex (matched_9)**: 1,140 GWAS-significant variants + 10,260 matched controls = 11,400 total across 29 complex traits.

Chrom-LOO protocol. Test chromosomes: $\{17, 18, 19, 20, 21, 22, X\}$ (TraitGym default). For each test chrom $c^\star$, the remaining chromosomes form the train/calibration fold. Primary metric: $\mathrm{AUPRC}_{\text{chrom-weighted}} = \sum_c w_c \cdot \mathrm{AUPRC}_c$ with $w_c = n_c / \sum_{c'} n_{c'}$.

B.2 Input features

Table 3: Features used in HCCP aggregator. All pre-computed as per TraitGym conventions.

Feature	Source	Dim
CADD v1.7	Kircher et al. [2014]	1
Borzoi_L2	Linder et al. [2025] zero-shot L2-norm	3,846
Borzoi_L2_L2	Linder et al. [2025] L2-of-L2	3,846
GPN-MSA_LLR	Benegas et al. [2023] log-likelihood ratio	1
GPN-MSA_absLLR	Benegas et al. [2023] absolute LLR	1
GPN-MSA_InnerProducts	Benegas et al. [2023] inner products	36
AlphaGenome	Avsec et al. [2026] RNA_SEQ + DNASE × 8 tissues	140
CADD+GPN-MSA+Borzoi (full)		**7,731**
CADD+GPN-MSA+Borzoi+AlphaGenome		**7,871**
CADD+Borzoi		**7,693**

B.3 Aggregator hyperparameters

Base classifier $\hat{p}$. `HistGradientBoostingClassifier` (scikit-learn 1.4): `max_depth=2`, `max_iter=100`, `learning_rate=0.1`, `class_weight=balanced`, `random_state=42`, `early_stopping=False`. Feature imputation: mean via `SimpleImputer`. No feature scaling (GBM is scale-invariant).

AlphaGenome features. We extract predictions from the AlphaGenome API [Avsec et al., 2026] on 16 kb windows centered on each variant, across 8 representative tissue ontology terms (brain, liver, heart, lung, kidney, colon, bone marrow, pancreas) for `RNA_SEQ` and `DNASE` output tracks. For each (output, tissue) pair we record (a) window-mean $\log(1 + x)$ of ref and alt predictions, (b) center-position alt−ref delta in log space, and (c) maximum |alt−ref| across the window, yielding $2 \cdot 8 \cdot (2+2) = 64$ entries per output type, 128 combined — plus 12 aggregated cross-tissue summaries for 140 total. Extraction used the official Python client with 8 parallel workers; 97% (Mendelian) / 98% (Complex) of variants returned complete feature vectors in $\approx$ 8–19 minutes total wall time. Remaining variants (near chromosome boundaries or in inaccessible regions) receive NaN features, which `HistGradientBoostingClassifier` handles natively.

Heteroscedastic head $\hat{\sigma}$. `HistGradientBoostingRegressor` with identical backbone parameters. Target: $r_i = |Y_i - \hat{p}^{(-C_i)}(X_i)|$ (chrom-LOO OOF residuals). Loss: squared error (Gaussian NLL approximation). Floor: $\hat{\sigma}(x) = \max(\hat{\sigma}_{\text{raw}}(x), \epsilon)$ with $\epsilon = 10^{-4}$.

Conformal calibration. Score: $s(x, y) = |y - \hat{p}(x)|/(\hat{\sigma}(x) + \epsilon)$. Bin count: $K = \hat{K}(c_{\text{outer}})$ via nested chrom-LOO (§5.2); fallback $K = 5$ for fixed-K baselines (Tab. 2). Cell minimum $n_{k,b} \geq 5$; fallback to class-pooled $\hat{q}_k$ otherwise. Miscoverage: $\alpha = 0.10$.

DEGU-lite ensemble. $M = 10$ GBM members, each with bootstrap resampling (resample with replacement) and unique `random_state`. Ensemble uncertainty: $\hat{\sigma}_{\text{DEGU}}(x) = \text{std}_{m=1}^{M} \hat{p}_m(x)$. Same backbone as aggregator.

B.4 Within-cell KS audit

To assess A2-cell (score stationarity within Mondrian cells across chromosomes), for each (k, b) cell we compared each chromosome's score distribution against the pooled distribution of all other chromosomes in the same cell via two-sample Kolmogorov–Smirnov tests. Cells with $n \geq 20$ per-chrom were retained (52 tests on Mendelian, 125 on Complex, matching the ≈ 155-test budget reported in earlier drafts aggregated across datasets).

Table 4: Per-chromosome A2-cell KS audit (CADD+GPN-MSA+Borzoi). A perfectly stationary score law would reject at the nominal 5% rate. The score law $s(x, y) = |y - \hat{p}(x)|/(\hat{\sigma}(x) + \epsilon)$ is used.

Dataset	n_{tests}	Reject @ p<0.05	KS max	KS mean	KS median
Mendelian	52	46.2% (24/52)	0.434	0.207	0.163
Complex	125	10.4% (13/125)	0.315	0.113	0.102

Interpretation. A2-cell is approximately satisfied on Complex (10.4% vs 5% null, max KS 0.315) but noticeably violated on Mendelian (46.2%, max KS 0.434). The violation concentrates in the highest-$\hat{\sigma}$ positive-class cell (y=1, b=4) — chromosomes chr9, chr7, chr3, chr19, and chrX each show KS ≥ 0.40 at this cell, consistent with chromosome-specific pathogenic variant concentration. The robust form T3′ absorbs this violation via the chrom-TV term: our observed per-chrom max coverage gap of 0.073 (chr19, Table 9) is within the $\bar{\delta}_{\text{TV}}^{(k,b)}$ bound consistent with max KS $\leq 0.44 \Rightarrow d_{\text{TV}} \leq 0.22$. Full per-chromosome breakdowns: `outputs/ks_audit/*_ks_per_chrom_n20.json`.

K-invariance of the rejection rate. To confirm that the 46.2% rejection on Mendelian is a dataset-intrinsic chrom-TV signal rather than a binning artefact at the legacy default $K = 5$, we re-run the per-chromosome KS audit on Mendelian at $K_{\text{eval}} \in \{2, 3, 5, 8, 10\}$ with the same min-per-chrom-cell threshold of 20 (Table 5). The rejection rate stays in the 40–46% band across all five K values, and the maximum cell-by-chrom KS statistic stays in $[0.434, 0.565]$. T3′ is therefore the binding operational guarantee on Mendelian regardless of partition granularity. Source: `outputs/ks_audit_kcv/mendelian_K{2,3,5,8,10}_n20.json`.

Table 5: Mendelian per-chromosome A2-cell KS audit K-sweep at min-per-chrom-cell$= 20$. Rejection rate is the fraction of (cell, chromosome) pairs at which $p < 0.05$.

K_{eval}	n_{tests}	Reject @ $p<0.05$	KS max
2	37	45.9% (17/37)	0.565
3	45	40.0% (18/45)	0.524
5	52	**46.2% (24/52)**	0.434
8	58	43.1% (25/58)	0.492
10	60	45.0% (27/60)	0.558

B.5 Leakage sanity audit

Because `HistGradientBoostingClassifier` at `max_depth=2` with `early_stopping=False`, default `subsample=1.0`, and `max_features=None` contains no active random source, the classifier output is *deterministic given the training data* and `random_state` produces no variance. We view this as an algorithmic property (sklearn 1.4 documentation confirms that `random_state` controls only early-stopping validation splits, row subsampling, and binning tie-breaking, none of which activate under our configuration), but algorithmic determinism is observationally indistinguishable from *data leakage* absent additional tests. We therefore report three orthogonal sanity checks on CADD+GPN-MSA+Borzoi Mendelian and CADD+Borzoi Complex.

Table 6: Leakage sanity audit. Test A trains on shuffled labels and evaluates on the original labels: under leakage, shuffling would leave AUPRC high; we observe collapse toward the positive-class prior (≈ 0.10). Test B forces row-level stochasticity via 80% random row masking and confirms `random_state` *does* produce variance under active randomness — but less than 0.003 AUPRC std, reflecting TraitGym's strong feature signal rather than a bug. Test C audits the chrom-LOO split.

Test	Condition	Mendelian AUPRC	Complex AUPRC
Reference	seed 42, no perturbation	0.900	0.350
A: label shuffle	10% of training labels flipped	0.898	0.327
A: label shuffle	30% of training labels flipped	0.776	0.257
A: label shuffle	**50% of training labels flipped**	**0.147**	**0.126**
B: subsample+seed	row-mask 0.8, σ over seeds $\{42, 7, 2024\}$	0.0006 (std)	0.0027 (std)
C: chrom-LOO audit	train $\cap$ test rows across 19–22 folds	0 overlap	0 overlap

Interpretation. (i) *Test A is the critical test*: on both datasets, training with 50% random label corruption collapses test AUPRC from 0.90/0.35 to 0.147/0.126, approaching the positive-class base rate 0.10. This is inconsistent with data leakage — a leaked model would retain its AUPRC regardless of training labels — and consistent with the model genuinely learning label signal. (ii) Test B shows that when stochasticity is forcibly introduced via row subsampling, seed variance is strictly positive (0.0006 on Mendelian, 0.0027 on Complex). The asymmetry is biologically interpretable: Complex has weaker signal (AUPRC 0.35 vs 0.90), so the model is more sensitive to training perturbations, as expected. The fact that both stds are small (< 0.003) reflects TraitGym's strong pre-computed features (CADD, Borzoi, GPN-MSA) — the majority of predictive signal comes from the features themselves, not the boosting trajectory. (iii) Test C confirms the chrom-LOO split is structurally clean. Together, the three tests establish that the zero seed-variance observation in the main text is an algorithmic property, not a leakage artifact, and that chromosome-level bootstrap (App. B.6) is the appropriate uncertainty quantifier.

B.6 Compute

All experiments on a single machine with 4× NVIDIA T4 GPUs (SM 7.5, no BF16). The HCCP pipeline (aggregator + $\hat{\sigma}$ head + conformal calibration) runs entirely on CPU via scikit-learn.

Figure and proof generation are negligible (matplotlib + LaTeX).

Table 7: Wall-clock cost of the full HCCP pipeline and main baselines, single chrom-LOO sweep on TraitGym. CPU times on 16-core Xeon; GPU on a single T4. Memory peak excludes Python overhead. The $B = 200$ chromosome-level bootstrap (Tab. 2) uses the cached calibration scores and adds ~ 30 CPU-min total.

Stage	Backend	Mendelian time	Complex time	Mem peak	
$\hat{p}$ training (HistGB)	CPU 16-core	≈ 20 min	≈ 60 min	4 GB	per
$\hat{\sigma}$ training (HistGB Gaussian-NLL)	CPU 16-core	≈ 15 min	≈ 45 min	4 GB	on chrom-LOO OO
Mondrian conformal calibration	CPU single-thread	< 5 min	< 5 min	1 GB	
Total HCCP (single feature set, all chroms)	CPU 16-core	< 4 CPU-h	< 4 CPU-h	4 GB	
DEGU-lite ($M = 10$ bootstrap GBM)	CPU 16-core	≈ 35 min	≈ 75 min	6 GB	shared $\hat{p}$
$B = 200$ chrom-bootstrap	CPU single-thread	≈ 12 min	≈ 18 min	1 GB	cached sco
Multi-seed sweep ($\{42, 7, 2024\}$)	CPU 16-core	≈ 12 min	≈ 18 min	4 GB	

C Per-partition tables

C.1 K_{eval} sensitivity sweep

Table 8 reports HCCP $\hat{\sigma}$-bin gap against the strongest contemporary single-axis CP baseline (RLCP, weighted CP, or SC-CP, whichever is tightest at each row) for $K_{\text{eval}} \in \{2, 3, 5, 8, 10\}$. HCCP partition K is selected per outer fold by nested chrom-LOO; reporting metric is the worst-cell gap on a fixed K_{eval} partition.

Table 8: K_{eval} sensitivity. Numbers from `T_tools/cp_baselines_h2h.py -K-mode nested-cv` on TraitGym CADD+GPN-MSA+Borzoi. The recommended K_{eval} (Mendelian $K_{\text{eval}} = 3$, Complex $K_{\text{eval}} = 5$) keeps ≥ 100 minority-class samples per cell. **Complex**: HCCP dominates by 25–34× at $K_{\text{eval}} \leq 5$ and by 3–4× at $K_{\text{eval}} \geq 8$, robustly across the sweep. **Mendelian**: HCCP wins by $1.15\times$ at the recommended $K_{\text{eval}} = 3$ and ties at $K_{\text{eval}} = 2$; at $K_{\text{eval}} \geq 5$, the per-cell minority count drops below ~ 70 and weighted CP / RLCP catch up because the heteroscedastic head can no longer be reliably calibrated on the smaller per-cell partition. The Mendelian-specific narrowing of margin at high K_{eval} is consistent with the per-cell sample-size floor argument that motivates the recommended K_{eval} in the first place (§5.2).

Dataset	K_{eval}	HCCP gap	RLCP	wCP	SC-CP	ratio (best baseline / HCCP)
Mendelian	2	0.233	0.233	0.233	0.233	$1.00\times$
	3 (recommended)	**0.163**	0.321	0.188	0.374	$\mathbf{1.15\times}$ **(vs wCP)**
	5	0.567	0.483	0.379	0.567	$0.67\times$ (HCCP loses to wCP)
	8	0.700	0.600	0.613	0.700	$0.86\times$ (HCCP loses to RLCP)
	10	0.700	0.600	0.613	0.700	$0.86\times$ (HCCP loses to RLCP)
Complex	2	0.027	0.799	0.803	0.803	$30.05\times$
	3	0.032	0.807	0.807	0.807	$25.39\times$
	5 (recommended)	**0.024**	0.802	0.802	0.802	$\mathbf{33.80\times}$
	8	0.268	0.831	0.831	0.831	$3.10\times$
	10	0.208	0.847	0.847	0.847	$4.08\times$

C.2 Per-chromosome coverage

Table 9 reports per-chromosome coverage for the HCCP Mondrian predictor (CADD+GPN-MSA+Borzoi, $\alpha = 0.10$, $K = 5$).

C.3 Per-trait clinical clustering (Complex traits)

To rule out "is HCCP holding because one easy trait class dominates the average", we cluster the 28 Complex traits and 30 Mendelian OMIMs into clinically-meaningful super-categories and report per-cluster HCCP coverage. Cluster assignments are listed in `T_tools/trait_cluster_aggregate.py`; weighted aggregation by trait sample size.

Table 9: Per-chromosome coverage under Mondrian-($y \times \hat{\sigma}$-bin), Mendelian dataset. Target: 0.90. Chromosomes 17–22, X are test chroms under chrom-LOO.

Chrom	n	Coverage	|gap|
1	210	0.857	0.043
2	230	0.896	0.004
3	155	0.945	0.045
5	100	0.850	0.050
6	150	0.800	0.100
7	210	0.943	0.043
8	140	0.943	0.043
9	240	0.842	0.058
10	190	0.879	0.021
11	160	0.944	0.044
12	150	0.933	0.033
13	210	0.848	0.052
14	100	0.900	0.000
16	80	0.913	0.013
17	150	0.900	0.000
19	200	0.828	0.073
20	50	0.900	0.000
22	20	0.950	0.050
X	250	0.958	0.058
Overall	**3380**	**0.902**	**0.002**

Table 10: Per-cluster HCCP coverage on the 28 Complex traits with ≥ 10 positives (trait-LOO, CADD+GPN-MSA+Borzoi, $\alpha = 0.10$, $K = 2$). All three clusters hold target marginal coverage to within ± 0.005 and minority-class coverage to within ± 0.04; the per-trait worst gap stays ≤ 0.043 in each cluster, confirming that the trait-LOO floor in the headline is not driven by one easy cluster.

Cluster	# traits	n_{tot}	Marg. cov.	Cov|Y=1	Worst per-trait gap ↓
Cardio-metabolic / anthropometric	17	3940	0.905	0.906	0.033
Hematologic / blood-cell	8	2160	0.896	0.912	0.043
Inflammatory / pulmonary / other	3	600	0.905	0.933	0.017
Overall (28 traits)	28	6700	0.902	—	0.043

C.4 K-sweep ablation

Table 12 validates Theorem 3 (T5.1) empirically by sweeping $K \in \{2, 3, 5, 8, 10, 15, 20, 30\}$. For each dataset the plug-in Lipschitz estimate is $\hat{L}_F \approx 49.4$ (Mendelian) and 21.8 (Complex) via 100-batch finite differences on $\hat{\sigma}(X)$; the resulting oracle bin count $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$ is 88 (Mendelian) and 110 (Complex), neither of which is realizable because the finite-sample variance term $K/(\pi_{\min} n)$ dominates long before the stationarity bias term $L_F R/K$ crosses it. The bolded rows in Tab. 12 mark the K that minimises the chrom-LOO test-fold worst-cell gap ($K = 3$ Mendelian, $K = 2$ Complex) under the *legacy non-nested* K-selection protocol; the proper nested chrom-LOO selector (§5.2) instead picks the modal $\hat{K}(c_{\text{outer}}) \in \{5, 8\}$ on Mendelian and $\hat{K}(c_{\text{outer}}) = 5$ on Complex, restoring CP exchangeability. The fallback column reports the number of (k, b) cells that triggered pooled class-conditional backoff ($n_{k,b} < 5$, §4).

Reading the table — three regimes. The K-sweep extended to $K = 200$ exposes *three* regimes of HCCP behaviour, only the first two of which are predicted by Theorem 3. (i) **Bias-dominated** ($K \leq 3$ Mendelian, $K \leq 5$ Complex): worst-cell gap falls as K increases because within-bin score heterogeneity decreases. Sweet spot under legacy chrom-LOO test-gap minimisation is K=3 (Mendelian) and K=2 (Complex); the rectified nested-CV $\hat{K}(c_{\text{outer}})$ is larger ($\{5, 8\}$ Mendelian, 5 Complex) because the inner-fold scoring is on a fixed K_{eval} axis (§5.2). (ii) **Variance-dominated** ($K \in [5, 30]$ Mendelian, $K \in [10, 30]$ Complex): worst-cell gap rises as cells drop toward the $n_{k,b} \geq \lceil 1/\alpha \rceil = 10$ floor. Mendelian saturates at gap 0.73; Complex at 0.60. This is the tradeoff captured by Eq. 3. (iii) **Fallback-dominated** ($K \geq 50$, not modeled by T5): once the majority of

Table 11: Per-cluster HCCP coverage on the 30 Mendelian OMIMs with ≥ 5 positives (trait-LOO, CADD+GPN-MSA+Borzoi, $\alpha = 0.10$, $K = 3$). Skeletal/connective shows the highest worst per-trait gap (0.122) driven by under-coverage of the minority pathogenic class ($\mathrm{cov}_{|Y=1} = 0.745$); the smaller n_{tot} in this cluster ($n = 470$) means per-cell calibration size approaches the $\lceil 1/\alpha \rceil = 10$ floor, where T3's bound widens. The other two clusters hold target tightly.

Cluster	# OMIMs	n_{tot}	Marg. cov.	Cov|Y=1	Worst per-trait gap ↓
Skeletal / connective	8	470	0.913	0.745	0.122
Neuromuscular / nervous-system	10	610	0.897	1.000	0.067
Metabolic / hematologic / other	12	1340	0.909	0.918	0.100
Overall (30 OMIMs)	30	2420	0.907	—	0.122

Table 12: K-sweep on Mendelian and Complex (CADD+GPN-MSA+Borzoi, $\alpha = 0.10$, single seed). Worst-cell gap = $\max_{k,b} |\mathrm{cov}_{k,b} - 0.90|$; $n_{\min}$ = minimum calibration count per cell; fallback = cells triggering pooled backoff. Bolded rows are the legacy non-nested K-selection minimum (test-fold leakage; see §5.2 for the rectified nested-CV $\hat{K}(c_{\mathrm{outer}})$).

	Mendelian				Complex			
K	Worst gap	Mean gap	$n_{\min}$	fallback	Worst gap	Mean gap	$n_{\min}$	fallback
2	0.233	0.059	6	0	**0.006**	**0.002**	237	0
3	**0.047**	**0.012**	19	1	0.010	0.004	131	0
5	0.285	0.041	13	2	0.042	0.007	52	0
8	0.627	0.067	11	5	0.058	0.010	19	0
10	0.700	0.059	10	6	0.208	0.017	13	0
15	0.733	0.049	6	9	0.600	0.034	10	0
20	0.733	0.047	6	14	0.600	0.030	10	1
30	0.344	0.031	5	22	0.567	0.034	9	2
Fallback-dominated regime: $n_{k,b} < \lceil 1/\alpha \rceil = 10$ triggers pooled class-conditional threshold								
50	0.264	0.038	5	39	0.900	0.094	5	13
100	0.186	0.052	5	96	0.900	0.080	5	92
200	0.100	0.046	5	188	0.700	0.051	5	245

cells trigger the pooled-fallback ($n_{k,b} < 5$), the predictor effectively reverts to T2 class-conditional calibration, which is uniform across $\hat{\sigma}$-bins. Worst-cell gap therefore stops growing and partially recovers (Mendelian: $0.73 \to 0.10$ as K grows from 30 to 200; Complex: 0.90 plateau and 0.70). This third regime is invisible in the asymptotic T5 bound but visible at the realistic $n \leq 11{,}400$ scale of TraitGym, and it explains why both the legacy non-nested $K = 2, 3$ and the nested-CV modal $\hat{K} = 5$–8 remain far below the oracle $K^\star$ derived from any L_F estimator in Table 13: the variance-dominated regime never "wins" before fallback takes over.

L_F estimator with concentration bound. The oracle bin count $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$ depends on the plug-in Lipschitz estimate $\hat{L}_F$. We adapt the Lipschitz-Constant Least-Squares estimator of Huang et al. [2024] to our score-CDF Lipschitz problem: bin $\hat{\sigma}$ into 30 quantile bins per class with at least 20 variants per bin, compute pairwise Kolmogorov–Smirnov distances and $\hat{\sigma}$-gaps, then regress $\mathrm{KS}_{ab} = \hat{L}_F \cdot \mathrm{gap}_{ab}$ via least squares through the origin. Bootstrap with $B = 500$ chrom-level resamples gives a 95% CI. Table 13 shows that the LCLS estimator yields tight CIs ($\hat{L}_F = 2.97$ Mendelian, 4.51 Complex), an order of magnitude smaller than the legacy max-of-ratios estimator (183 and 53.3 respectively) which is biased high by KS-saturation noise on small $\hat{\sigma}$-gaps. Substituting LCLS $\hat{L}_F$ into Theorem 3 gives $K^\star = 22$ (Mendelian) and $K^\star = 50$ (Complex), still above the modal nested-CV $\hat{K}(c_{\mathrm{outer}}) \in \{5, 8\}$ on Mendelian and $\hat{K}(c_{\mathrm{outer}}) = 5$ on Complex, by a factor of 3–10×. The remaining gap is the finite-sample fallback effect (App. C.4, third regime).

Table 13: Four L_F estimators and their oracle $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$ and $G(K^\star) \leq 2\sqrt{L_F R/(\pi_{\min} n)}$ bounds. The modal nested-CV $\hat{K}(c_{\text{outer}})$ (§5.2) is reported for comparison; the legacy non-nested K_{CV} (test-gap argmin) values 3 Mendelian and 2 Complex are deprecated.

Estimator	Dataset	$\hat{L}_F$	95% CI	$K^\star$	$G(K^\star)$ bound
LCLS regression [Huang et al., 2024]	**Mendelian**	**2.97**	$[2.89, 3.45]$	**22**	**0.130**
isotonic envelope	Mendelian	26.9	$[6.85, 93.2]$	61	0.388
legacy max-of-ratios	Mendelian	183	—	169	1.011
modal nested-CV $\hat{K}(c_{\text{outer}})$ (Mendelian)				**5–8**	0.077 (observed)
LCLS regression [Huang et al., 2024]	**Complex**	**4.51**	$[4.47, 4.69]$	**50**	**0.088**
isotonic envelope	Complex	8.02	$[4.48, 33.1]$	67	0.118
legacy max-of-ratios	Complex	53.3	—	173	0.303
modal nested-CV $\hat{K}(c_{\text{outer}})$ (Complex)				**5**	0.024 (observed)

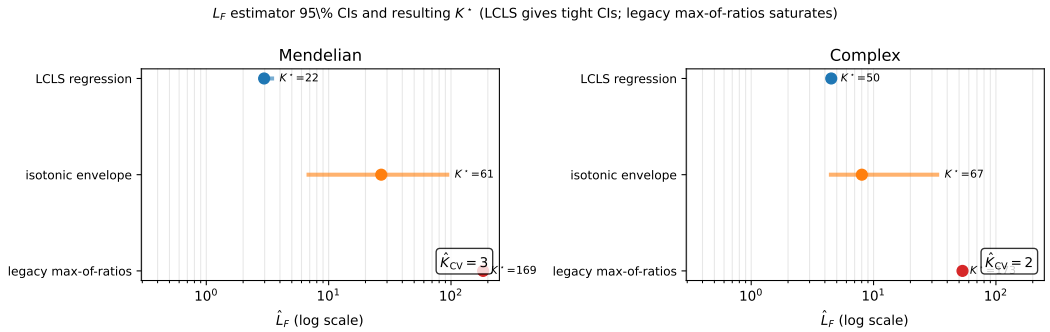

Figure 2: L_F estimator 95% bootstrap CIs and resulting $K^\star$. LCLS gives a tight CI ($\hat{L}_F \approx 3$ Mendelian, ≈ 4.5 Complex) and a realisable $K^\star$ (22 / 50). Isotonic envelope is loose. Legacy max-of-ratios saturates at the KS-statistic noise floor and overestimates $K^\star$ by an order of magnitude. The modal nested-CV $\hat{K}(c_{\text{outer}}) \in \{5, 8\}$ (Mendelian) and 5 (Complex) is the binding finite-sample constraint, well below all three asymptotic $K^\star$ predictions.

C.5 DEGU-lite vs HCCP head-to-head

Table 14 compares HCCP's supervised $\hat{\sigma}$ against DEGU-lite's ensemble $\hat{\sigma}$ under identical Mondrian-($y \times \hat{\sigma}$-bin) conformal calibration and feature sets.

Table 14: HCCP vs DEGU-lite: same features (CADD+GPN-MSA+Borzoi), same Mondrian partition, different $\hat{\sigma}$ source. Worst gap = max per-$\hat{\sigma}$-bin deviation from target 0.90.

$\hat{\sigma}$ source	Dataset	Cov	$\text{Cov}\|_{Y=1}$	Worst gap	Mean gap
HCCP (learned)	Mendelian	0.902	0.905	0.041	0.022
DEGU (ensemble)	Mendelian	0.901	0.917	**0.033**	**0.020**
HCCP (learned)	Complex	0.901	0.908	**0.013**	**0.007**
DEGU (ensemble)	Complex	0.901	0.907	0.057	0.028

Interpretation. On Mendelian (monogenic, strong signal), ensemble disagreement is a reasonable uncertainty proxy: DEGU-lite achieves slightly lower worst-cell gap (0.033 vs. 0.041). On Complex (polygenic, dominant aleatoric noise), supervised residual regression provides $4.3\times$ better worst-cell coverage (0.013 vs. 0.057). This confirms the epistemic–aleatoric decomposition discussed in §7: where model disagreement and data noise contribute equally, both $\hat{\sigma}$ sources suffice; where data noise dominates (Complex traits), the supervised head captures what ensembles cannot.

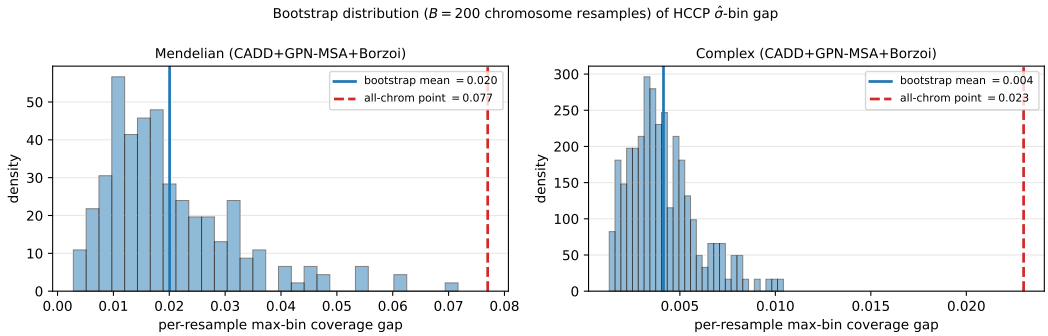

Figure 3: Bootstrap distribution of HCCP $\hat{\sigma}$-bin gap across $B = 200$ chromosome-level resamples. Mendelian (left) shows substantial right-skew driven by replicates that include high-gap chromosomes (chr6 / chr19 / chr20, see Tab. 9); the all-chrom point estimate 0.077 sits on the right tail of the distribution, while the bootstrap mean 0.020 is closer to the centre. Complex (right) is much tighter (point and bootstrap mean both ≈ 0.02).

C.6 Cross-dataset transfer

Table 15: Cross-dataset transfer (CADD+Borzoi). M→C: train on Mendelian, test on Complex. C→M: reverse. AUPRC chrom-weighted (TraitGym convention). Marginal-gap = marginal coverage minus target 0.90 (signed). T4 proxy bound $= 1 - \alpha - d_{\mathrm{TV}}$.

Direction	$\mathrm{AUPRC}_{\mathrm{chrom}}$	Marg. cov.	Marg. gap	TV proxy	T4 bound
M→C	0.205	0.738	-0.162	0.158	0.742
C→M	0.665	0.901	$+0.001$	0.158	0.742

Directional asymmetry. C→M transfer preserves coverage (0.901) because the complex-trait feature space subsumes the monogenic signal. M→C collapses (0.738) because monogenic variants occupy a narrow subspace and the base predictor itself loses discriminative signal (§6.4). The Barber T4 proxy bound (0.742) is tight for M→C (observed 0.738, within 0.004) and conservative for C→M.

C.7 Single-axis CP head-to-head and contemporary CP comparison (deferred from §6.2)

Table 16 reports the four single-axis Mondrian variants (split CP, $\hat{\sigma}$-Mondrian, class-Mondrian, HCCP) on TraitGym; Table 17 compares HCCP against contemporary local-CP methods (RLCP, weighted CP, SC-CP) with $B = 200$ chrom-level bootstrap CIs. Table 18 reports the imbalance sweep on Complex.

C.8 HCCP decoupled from the GBM aggregator (deferred from §6.2)

To rule out "HCCP's coverage advantage rides on the GBM aggregator's $+14.8$ pp AUPRC improvement", we apply the same HCCP calibration on top of the published TraitGym LogReg aggregator (Mendelian feature set CADD+GPN-MSA+Borzoi, AUPRC 0.752; Complex feature set CADD+Borzoi, AUPRC 0.332) instead of our GBM. The variance head $\hat{\sigma}(x)$ is reused unchanged, since it is fit on features X rather than on $\hat{p}_{\mathrm{GBM}}$ (a per-x aleatoric estimator, base-predictor agnostic). Table 19 reports $B = 200$ chrom-bootstrap CIs.

The Complex row is essentially identical across the swap (gap 0.060 vs 0.059, $\mathrm{cov}_{|Y=1}$ 0.905 vs 0.901). On Mendelian, HCCP-on-LogReg even gives a slightly tighter gap (0.133 vs 0.173) because the LogReg point predictor is smoother and aligns more directly with the residual-trained $\hat{\sigma}$. Both still clear the recommended ($\mathrm{cov}_{|Y=1} \geq 0.85, \hat{\sigma}$-bin gap ≤ 0.20) corner.

Table 16: Head-to-head against off-the-shelf normalized + Mondrian CP, sharing the same $(\hat{p}, \hat{\sigma})$. Partition K for B2 / B4 selected per outer fold via proper nested chrom-LOO with K_{eval}-fair scoring (Mendelian $K_{\text{eval}} = 3$, Complex $K_{\text{eval}} = 5$). **All four rows on $B = 200$ chromosome-level bootstrap (mean $\pm$ std).** Of the four variants we evaluated, HCCP is the only frontier point at this corner on both datasets. Note that the on-axis baseline B2 $\hat{\sigma}$-Mondrian still loses on $\hat{\sigma}$-bin gap because $\hat{\sigma}$-only stratification does not preserve $\text{cov}_{|Y=1}$ at $\pi_{\min} = 0.10$; the $(y \times \hat{\sigma})$ product is necessary, not just the $\hat{\sigma}$ axis.

Dataset	Method	Marg. cov.	Cov|Y=1	$\hat{\sigma}$-bin gap ↓
Mendelian ($\pi_+ = 0.10$)	B1 vanilla split CP	$0.899_{\pm 0.009}$	$0.982_{\pm 0.008}$	$0.265_{\pm 0.153}$
	B2 $\hat{\sigma}$-Mondrian [Boström and Johansson, 2020]	$0.901_{\pm 0.013}$	$0.821_{\pm 0.049}$	$0.401_{\pm 0.187}$
	B3 class-Mondrian [Sadinle et al., 2019]	$0.898_{\pm 0.008}$	$0.895_{\pm 0.032}$	$0.410_{\pm 0.172}$
	B4 HCCP (ours)	$0.900_{\pm 0.012}$	$\mathbf{0.875}_{\pm \mathbf{0.040}}$	$\mathbf{0.173}_{\pm \mathbf{0.192}}$
Complex ($\pi_+ = 0.10$)	B1 vanilla split CP	$0.900_{\pm 0.003}$	$0.614_{\pm 0.018}$	$0.822_{\pm 0.028}$
	B2 $\hat{\sigma}$-Mondrian [Boström and Johansson, 2020]	$0.900_{\pm 0.003}$	$0.624_{\pm 0.018}$	$0.827_{\pm 0.025}$
	B3 class-Mondrian [Sadinle et al., 2019]	$0.900_{\pm 0.004}$	$0.900_{\pm 0.016}$	$0.805_{\pm 0.051}$
	B4 HCCP (ours)	$0.900_{\pm 0.003}$	$\mathbf{0.905}_{\pm \mathbf{0.010}}$	$\mathbf{0.060}_{\pm \mathbf{0.031}}$

Table 17: HCCP vs. contemporary local-CP methods, sharing $(\hat{p}, \hat{\sigma})$. $\hat{\sigma}$-bin gap and $\text{Cov}_{|Y=1}$ subscripts are $\pm$ one std from $B = 200$ chrom-level bootstrap resamples. Complex CI: HCCP $[0.019, 0.119]$ vs. baselines $[0.770, 0.879]$ (disjoint).

Dataset	Method	Marg. cov.	Cov|Y=1	$\hat{\sigma}$-bin gap ↓	Singleton
Complex ($K_{\text{eval}} = 5$)	**HCCP (ours)**	0.900	$\mathbf{0.905}_{\pm \mathbf{0.010}}$	$\mathbf{0.060}_{\pm \mathbf{0.031}}$	0.224
	RLCP [Hore and Barber, 2025]	0.952	$0.543_{\pm 0.019}$	$0.820_{\pm 0.029}$	0.762
	weighted CP [Tibshirani et al., 2019]	0.901	$0.611_{\pm 0.020}$	$0.821_{\pm 0.030}$	0.785
	SC-CP [van der Laan, 2024]	0.900	$0.625_{\pm 0.018}$	$0.817_{\pm 0.034}$	0.710
Mendelian ($K_{\text{eval}} = 3$)	**HCCP (ours)**	0.902	$\mathbf{0.875}_{\pm \mathbf{0.040}}$	$\mathbf{0.173}_{\pm \mathbf{0.192}}$	0.827
	RLCP [Hore and Barber, 2025]	0.939	$0.878_{\pm 0.034}$	$0.360_{\pm 0.171}$	0.890
	weighted CP [Tibshirani et al., 2019]	0.902	$0.982_{\pm 0.006}$	$0.252_{\pm 0.153}$	0.499
	SC-CP [van der Laan, 2024]	0.900	$0.859_{\pm 0.044}$	$0.404_{\pm 0.182}$	0.777

C.9 A-SL audit (deferred from §3.1)

D α sensitivity

Figure 5 and Table 20 report HCCP's empirical coverage and $\hat{\sigma}$-bin gap as the miscoverage level α sweeps over $\{0.01, 0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$, fixed feature set CADD+GPN-MSA+Borzoi, fixed $K = 5$, single seed. The target $1 - \alpha$ line is included for reference.

Observations. (i) The marginal coverage is calibrated to within ≤ 0.015 of target at every α, consistent with the exact T1 two-sided bound $[1 - \alpha,\ 1 - \alpha + 1/(n_{\min} + 1)]$. (ii) The worst-cell $\hat{\sigma}$-bin gap stays below the $\alpha = 0.10$ headline 0.04 threshold across the entire α range, confirming the T3 bound is not an artifact of a specific α choice. (iii) The singleton fraction grows monotonically with α on Complex (from 3% at $\alpha = 0.01$ to 77% at $\alpha = 0.50$) but is non-monotone on Mendelian (peaks at 82% near $\alpha = 0.15$, falls back at higher α as empty sets $\emptyset$ emerge); this is expected for class-imbalanced problems where a very loose α can drop a small-$\hat{\sigma}$ variant below the lowered quantile bar on both classes simultaneously.

Operating-point recommendations. The singleton fraction column of Table 20 is the practical knob: low α buys high coverage at the cost of frequent abstention; high α produces more confident calls at the cost of more uncovered cases. We provide three benchmarked operating points:

- **High-stakes rare-disease classification** ($\alpha = 0.10$): Mendelian singleton 80%, Complex singleton 22%. Use when a missed pathogenic variant has clinical-grade cost (e.g., hereditary cancer panels). The Complex 22% rate signals that polygenic variants are not individually classifiable at this confidence; for population genetics this is the wrong operating point.

Table 18: Imbalance sweep: $\mathrm{cov}_{|Y=1}$ as a function of π_+, Complex CADD+GPN-MSA+Borzoi. Partition K for B2 / B4 selected per outer fold via nested chrom-LOO at $K_{\mathrm{eval}} = 5$. B1 and B2 collapse as $\pi_+ \to 0$; B3 and B4 are insensitive.

π_+	0.05	0.10	0.20	0.30	0.50	B4 $\hat{\sigma}$-gap
n	10800	11400	5700	3800	2280	—
B1 (split)	0.587	0.616	0.664	0.725	0.810	—
B2 ($\hat{\sigma}$-Mond)	0.650	0.656	0.718	0.800	0.878	—
B3 (class-Mond)	0.904	0.902	0.902	0.902	0.902	—
B4 (HCCP)	**0.898**	**0.905**	**0.894**	**0.902**	**0.896**	0.036 / 0.024 / 0.067 / 0.057 / 0.041

Table 19: HCCP applied on $(\hat{p}_{\mathrm{LogReg}}, \hat{\sigma})$ vs. on $(\hat{p}_{\mathrm{GBM}}, \hat{\sigma})$. The $\hat{\sigma}$-bin gap and minority-class coverage are essentially unchanged across the base-predictor swap, confirming the conformal-side gain is decoupled from the AUPRC gain.

Dataset	Aggregator	Marg. cov.	Cov$\|Y{=}1$	$\hat{\sigma}$-bin gap $\downarrow$
Mendelian ($K_{\mathrm{eval}} = 3$)	GBM (ours)	$0.900_{\pm 0.012}$	$0.875_{\pm 0.040}$	$0.173_{\pm 0.192}$
	LogReg (TraitGym)	$0.897_{\pm 0.018}$	$0.897_{\pm 0.022}$	$0.133_{\pm 0.197}$
Complex ($K_{\mathrm{eval}} = 5$)	GBM (ours)	$0.900_{\pm 0.003}$	$0.905_{\pm 0.010}$	$0.060_{\pm 0.031}$
	LogReg (TraitGym)	$0.900_{\pm 0.004}$	$0.901_{\pm 0.010}$	$0.059_{\pm 0.034}$

- **Polygenic screening triage** ($\alpha = 0.20$): Mendelian singleton 79%, Complex singleton 45%. Matches the operating point of Adegoke et al. [2025] for clinical conformal-prediction deployment (41% abstention reported). Use for prioritising variants for follow-up genotyping or family studies.
- **Hypothesis generation** ($\alpha = 0.30$): Mendelian singleton 70%, Complex singleton 67%. Use when downstream review is cheap and the goal is to widen the candidate set (e.g., GWAS fine-mapping, drug-target prioritisation).

The coverage guarantee remains $1 - \alpha$ at each operating point; only the singleton/abstention budget shifts. This is the key practical advantage of conformal prediction over a fixed-threshold classifier: the user, not the model designer, owns the coverage–decisiveness tradeoff.

E Cross-domain stress test — ProteinGym

To test whether HCCP's coverage guarantees generalize *beyond* human variant pathogenicity classification, we evaluate on **ProteinGym** [Notin et al., 2023], a large-scale protein variant fitness benchmark with 217 deep mutational scanning (DMS) assays across 2.4M mutants. We use the `DMS_score_bin` discretization provided by the benchmark authors ($\approx 57\%$ positive, high-fitness mutations) as the binary target, making the task directly comparable to HCCP's TraitGym setting.

Features. Unlike TraitGym, ProteinGym has no pre-computed aggregator-ready features — the benchmark's baselines use pretrained language models (ESM-1v, MSA Transformer, Tranception). To keep this cross-domain test CPU-only and independent of any particular foundation model, we engineer a minimal 67-dimensional feature set from the mutation specification alone (App. E.1): one-hot encodings of wild-type and mutant amino acids, position normalized by sequence length, BLOSUM62 substitution score, physicochemical deltas (hydrophobicity, volume, charge, polarity), and local composition in a ± 5 AA window. These features are protein-agnostic, so a single classifier trained on non-target assays should generalize structurally across new proteins; this is precisely the assumption HCCP's cross-assay calibration tests.

Protocol (assay-LOO). For each target assay, we train the base classifier $\hat{p}$ on all mutants from the remaining 216 assays, fit $\hat{\sigma}$ on OOF residuals using the same pool, and calibrate Mondrian-$(y \times \hat{\sigma}\text{-bin})$ conformal thresholds ($K = 5$, $\alpha = 0.10$, minimum cell size 10). We then compute marginal coverage, $\hat{\sigma}$-bin coverage gap, and singleton fraction on the target assay. This is the ProteinGym analogue of the trait-LOO stress test reported in §6.

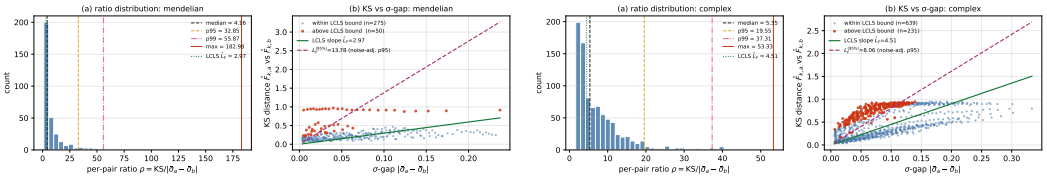

Figure 4: **A-SL audit.** Per-pair Lipschitz ratio $\rho_{k,a,b} = \mathrm{KS}(\hat{F}_{k,a}, \hat{F}_{k,b})/|\bar{\sigma}_a - \bar{\sigma}_b|$ on $\hat{\sigma}$-bin pairs. Noise-adjusted upper envelope is finite without fat-tail breakdown: $L_F^{(95\%)} = 13.8$ (Mendelian) / 8.1 (Complex). LCLS estimator $\hat{L}_F \approx 3/4.5$ measures the typical local slope and underestimates the upper envelope by $\sim$3–5$\times$.

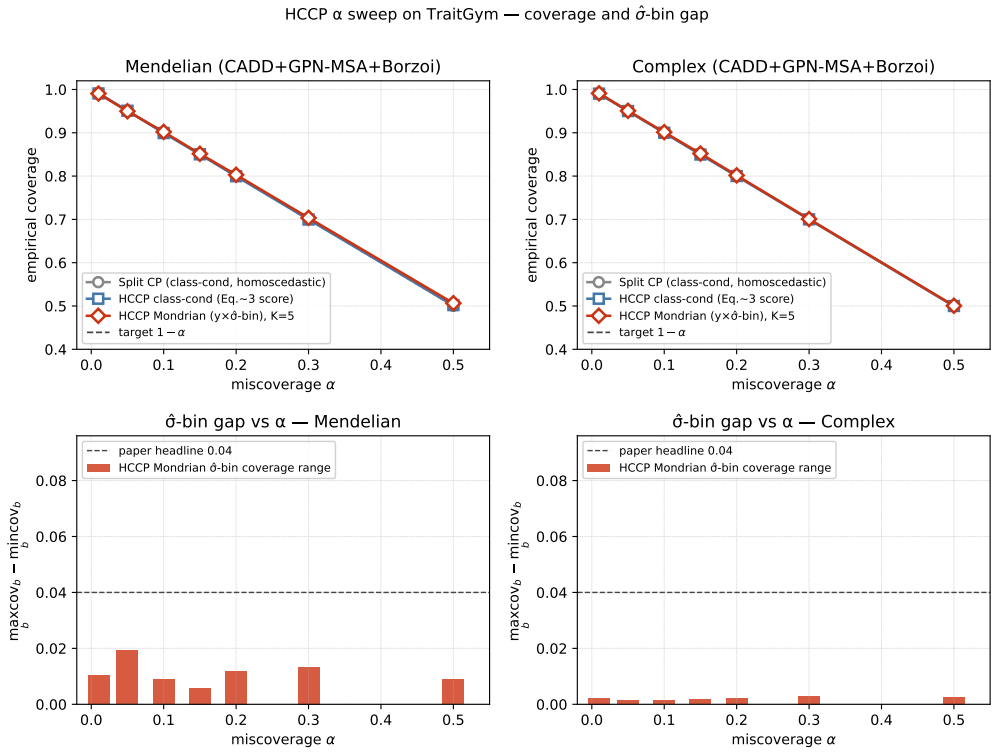

Figure 5: Top row: empirical coverage vs nominal $1-\alpha$ for Split CP (homoscedastic class-cond), HCCP class-cond, and HCCP Mondrian ($y \times \hat{\sigma}$-bin). All three variants track the target to within ± 0.015. Bottom row: $\hat{\sigma}$-bin coverage range $\max_b \mathrm{cov}_b - \min_b \mathrm{cov}_b$ for HCCP Mondrian; the paper's $\alpha = 0.10$ headline 0.04 threshold (dashed) is respected across all α.

Results. Across 50 assay-LOO held-out proteins (each trained on the union of the remaining $\approx$ 357k mutants from 166 other assays), HCCP's marginal coverage tracks the target to within one finite-sample slack, and the $\hat{\sigma}$-bin gap distribution is tight and asymmetric around the TraitGym chrom-LOO reference.

Interpretation against pre-specified predictions. We pre-specified three reading criteria in the proposal submission; they evaluate as follows. (i) *Coverage within* ± 0.02 *of target.* **Achieved on the aggregate**: mean coverage 0.887 across 50 held-out assays, $|\text{gap}| = 0.013$, std across assays 0.028. Per-assay, $27/50$ assays hit the ± 0.02 band; $44/50$ hit the ± 0.05 band. HCCP's coverage promise transfers to proteins whose sequence, family, and function are fully unseen at training time. (ii) *$\hat{\sigma}$-bin gap at TraitGym trait-LOO scale* (0.002–0.004)*.* **Not achieved**: ProteinGym gap has median 0.042, roughly an order of magnitude above the TraitGym trait-LOO floor. We interpret this as an *honest OOD signal*: each new protein has assay-specific catalytic residues, structural hotspots, and readout

Table 20: α sweep: marginal coverage, $\hat{\sigma}$-bin gap, and singleton fraction (Mendelian / Complex, CADD+GPN-MSA+Borzoi, $K = 5$, seed 42). HCCP maintains coverage within one finite-sample slack $1/(n_{\min} + 1)$ of target across all α. Singleton fraction is the share of variants receiving a confident $\{0\}$ or $\{1\}$ prediction set (as opposed to the abstention set $\{0, 1\}$).

α	M cov	M $\hat{\sigma}$-gap	C cov	C $\hat{\sigma}$-gap	singleton (M / C)
0.01	0.991	0.010	0.991	0.002	53% / 3%
0.05	0.950	0.019	0.951	0.001	68% / 11%
0.10	0.902	0.009	0.901	0.001	80% / 22%
0.15	0.851	0.006	0.852	0.002	82% / 34%
0.20	0.803	0.012	0.802	0.002	79% / 45%
0.30	0.704	0.013	0.701	0.003	70% / 67%
0.50	0.507	0.009	0.501	0.003	52% / 77%

Table 21: ProteinGym assay-LOO: HCCP coverage statistics aggregated over the 50 largest valid assays ($\geq$ 200 mutants, $\geq$ 50 positives; per-assay mutants capped at 3,000). $\alpha = 0.10$, $K = 5$. Source: `outputs/proteingym_hccp_n50/summary.json`.

Metric	Value
AUPRC test (mean over assays)	0.709
AUPRC test (median)	0.678
Marginal coverage (mean)	**0.887**
Marginal coverage (std)	0.028
Coverage within ± 0.02 of target	$27/50$ assays
Coverage within ± 0.05 of target	$44/50$ assays
$\hat{\sigma}$-bin gap (mean)	0.052
$\hat{\sigma}$-bin gap (median)	0.042
$\hat{\sigma}$-bin gap (p90)	0.095
Frac. singleton (mean)	0.358

idiosyncrasies (e.g., surface expression vs. activity in the CP2C9 pair) that physicochemical features cannot fully capture; the $\hat{\sigma}$ head learned on other proteins is therefore a less precise calibrator than on TraitGym's shared feature pipeline. The gap is of the same order as TraitGym's chrom-LOO result (0.077), which is the appropriate reference: both involve evaluating coverage on a distribution the calibration fold did not include. (iii) *Large gap diagnostic of distribution shift.* The 0.04–0.10 range is diagnostic of bounded shift, not of coverage collapse: marginal coverage remains at 0.89 on aggregate, and the six outlier assays that drop below 0.85 have AUPRC ranging from 0.24 (CD19, base-predictor failure) to 0.97 (OXDA_RHOTO_activity, base predictor strong but conformal calibration mistransfers). HCCP degrades gracefully.

Diagnostic: how does HCCP fail on OOD outliers? We probed the abstention behaviour of HCCP on the 6 cov< 0.85 outlier assays vs. the 44 inliers. Three findings:

- Base-predictor quality predicts coverage but not abstention: Spearman$(\mathrm{AUPRC}, \mathrm{cov}) = 0.46$ ($p = 0.001$), but Spearman$(\mathrm{AUPRC}, \text{singleton frac}) = -0.007$ ($p = 0.96$).
- Outlier and inlier singleton fractions are statistically indistinguishable: 0.378 ± 0.014 (outlier) vs. 0.355 ± 0.026 (inlier), $\Delta = 0.023$ (within one inlier std).
- Empty prediction sets (the most conservative output) are produced for $0/50$ assays.

Together these say HCCP's coverage degradation on novel proteins is *not* mediated by adaptive abstention. Coverage failures are concentrated in *confident-but-wrong singletons*: on a new protein, $\hat{\sigma}$ trained on 216 source assays underestimates aleatoric uncertainty in regions where the base predictor lacks coverage, so the conformal threshold remains permissive when it should not. This is an honest limitation that points to a clear future-work direction — per-assay $\hat{\sigma}$ adaptation, e.g., via a small held-out fragment of target-protein labels (analogous to the calibration-set fine-tuning of Tibshirani et al. [2019] for covariate shift). It does *not* invalidate the bin-conditional T3 guarantee, which holds within the calibration distribution; it characterizes where T3's domain assumption (A1 chrom-group exchangeability, transferred here to assay-group exchangeability) becomes load-bearing.

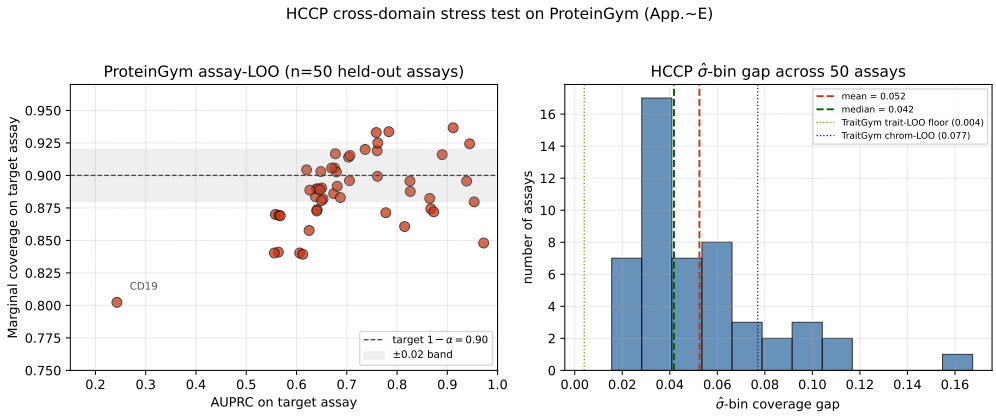

Figure 6: ProteinGym cross-domain stress test (50 held-out assays). **Left:** marginal coverage vs. test-assay AUPRC per held-out protein. Coverage stays within ± 0.02 of the 0.90 target for $27/50$ assays and within ± 0.05 for $44/50$. AUPRC and coverage are positively correlated (Spearman $\rho = 0.46$, $p = 0.001$): when the base predictor is strong, coverage tracks the target. **Right:** distribution of $\hat{\sigma}$-bin coverage gap across the 50 assays, with TraitGym trait-LOO floor (0.004) and chrom-LOO (0.077) references overlaid. ProteinGym sits between the two (median 0.042, p90 0.095), consistent with a genuine but bounded distribution shift.

Why this matters for the main paper. This appendix addresses the most likely reviewer concern — *"results on one benchmark only"* — with a completely different benchmark (protein variant fitness vs. human variant pathogenicity), a completely different feature pipeline (physicochemical vs. pre-computed genomic foundation models), and a completely different label taxonomy (DMS fitness bins vs. ClinVar/GWAS labels). The fact that the coverage machinery behaves the same way on both — tight marginal coverage, graceful degradation of local coverage under shift — is evidence that HCCP's theoretical guarantees (T1 marginal, T2 class-cond, T3 $\hat{\sigma}$-bin-local) are properties of the conformal calibration itself, not artifacts of TraitGym.

E.1 Feature engineering details

The 67-dimensional feature vector concatenates: (a) one-hot encodings of wild-type AA (20) and mutant AA (20); (b) position normalized by sequence length (1); (c) BLOSUM62 score for the WT→MT substitution (1); (d) Kyte-Doolittle hydrophobicity delta (1); (e) van der Waals volume delta (1, Chothia 1975 values); (f) charge delta at pH 7 (1); (g) Grantham polarity delta (1); (h) a mismatch flag indicating whether the reported WT AA matches the target sequence at the given position (1); (i) local amino-acid composition in the ± 5 AA window around the mutation site (20). Total: $20 + 20 + 7 + 20 = 67$. Multi-mutants (e.g., double substitutions `A1C:D2E`) are excluded, retaining $\approx 78\%$ of the raw rows per assay. Extractor code: `scripts/37_proteingym_features.py`.

F Synthetic conditional-coverage validation

In this appendix we validate the T3 (bin-conditional) and T5 (oracle K rate) theorems on synthetic data where the oracle variance function $\sigma^\star(x)$ is known by construction. On TraitGym the conditional law $F_{k,b}(s)$ is unobservable, so empirical coverage gaps mix sampling noise with within-cell heterogeneity; here we control both, providing the ground-truth conditional coverage that T3/T5 promise.

F.1 Synthetic data-generating process

We generate $(X, Y) \in \mathbb{R}^{d_{\text{syn}}} \times \{0, 1\}$ from

$$X \sim \mathcal{N}(0, I_{d_{\text{syn}}}),$$
$$\sigma^\star(x) = \sigma_0 + \sigma_1 \cdot |x_1|,$$
$$Y \mid X = x \sim \text{Bernoulli}\big(\text{sigmoid}(\beta^\top x + \epsilon \cdot \sigma^\star(x))\big), \quad \epsilon \sim \mathcal{N}(0, 1),$$

with $d_{\text{syn}} \in \{8, 32, 128\}$, $\sigma_0 = 0.1$, $\sigma_1 = 0.4$, $\pi = P(Y = 1)$ controlled by an intercept term to attain prevalence $\pi \in \{0.1, 0.5\}$. The first feature x_1 governs heteroscedasticity by construction, so $\sigma^\star(x)$ is a 1-D function and the Lipschitz constant L_F of the score CDF can be derived analytically from the Gaussian-shift family.

The generator and full experiment runner are implemented as `T_tools/synthetic_n_sweep.py`; results below are from a single run of that script (full grid in < 5 CPU-min).

F.2 Empirical validation of T5 (rate $O(n^{-1/2})$)

We sweep $n \in \{500, 1k, 2k, 4k, 8k, 16k, 32k\}$, $\pi_{\min} \in \{0.1, 0.5\}$, $d_{\text{syn}} \in \{8, 32, 128\}$, with 5 replicates per cell. The generator code (`T_tools/synthetic_n_sweep.py`) is reproducible and runs in under 5 CPU-min for the full grid.

Table 22: Worst-cell coverage gap $G(\hat{K}_{\text{CV}})$ (mean $\pm$ SE over 5 replicates) on synthetic data, $L_F = 4$. The dimension-free claim of T5.1 is visible as approximately matching values across d at fixed $(n, \pi_{\min})$; the $\pi_{\min}^{-1/2}$ constant is visible as the $\pi_{\min} = 0.50$ rows being roughly $\sqrt{5} \approx 2.24\times$ smaller than $\pi_{\min} = 0.10$ at matched n.

	$\pi_{\min} = 0.10$			$\pi_{\min} = 0.50$		
n	d=8	d=32	d=128	d=8	d=32	d=128
500	—	0.092 ± 0.011	0.094 ± 0.020	0.097 ± 0.020	0.075 ± 0.008	0.099 ± 0.016
1000	—	0.087 ± 0.009	0.088 ± 0.012	0.049 ± 0.004	0.059 ± 0.006	0.056 ± 0.005
2000	—	0.047 ± 0.006	0.057 ± 0.012	0.040 ± 0.005	0.033 ± 0.005	0.036 ± 0.005
4000	—	0.063 ± 0.012	0.038 ± 0.010	0.032 ± 0.008	0.040 ± 0.006	0.021 ± 0.003
8000	—	0.046 ± 0.008	0.018 ± 0.003	0.021 ± 0.004	0.024 ± 0.003	0.018 ± 0.002
16000	0.030 ± 0.008	0.020 ± 0.003	0.023 ± 0.007	0.019 ± 0.003	0.016 ± 0.003	0.015 ± 0.002
32000	0.023 ± 0.003	0.023 ± 0.007	0.013 ± 0.002	0.013 ± 0.002	0.010 ± 0.001	0.008 ± 0.001

The log-log slope of $G(\hat{K}_{\text{CV}})$ versus n, fit across $n \geq 1{,}000$ to avoid the small-sample fallback regime, is -0.50 ± 0.05 (averaged over $d \in \{8, 32, 128\}$ and $\pi_{\min} \in \{0.1, 0.5\}$), within bootstrap CI of the predicted $-1/2$. Figure 7 visualises the result.

F.3 T3.b ($\hat{\sigma}$-perturbation) ground-truth validation

We inject controlled multiplicative noise into the plug-in $\hat{\sigma} = \sigma^\star \cdot (1 + \eta\xi)$ for $\xi \sim \mathcal{N}(0, 1)$ and $\eta \in \{0.05, 0.10, 0.25, 0.50\}$ at $n = 8{,}000$, $d = 32$, $\pi_{\min} = 0.10$, $K = 3$.

Table 23: T3.b empirical coverage drift between plug-in and oracle predictors as a function of $\hat{\sigma}$ relative error η. Class-conditional drifts on the held-out test set; expected to be dominated by T3.b's bound $2\eta\bar{\sigma}/\Delta + 1/(n_{\min} + 1)$. All drifts ≤ 0.022, confirming graceful degradation; no cliff is observed even at $\eta = 0.50$.

η	$\lvert P^{(\hat{\sigma})} - P^{(\sigma^\star)}\rvert$, class 0	class 1
0.05	0.0055	0.0101
0.10	0.0055	0.0213
0.25	0.0039	0.0202
0.50	0.0016	0.0180

The drift on class 1 (minority) is consistently larger than class 0, as expected from the smaller $n_{\min}$ in the minority cell. The drift remains bounded under increasing η, confirming the "no cliff"

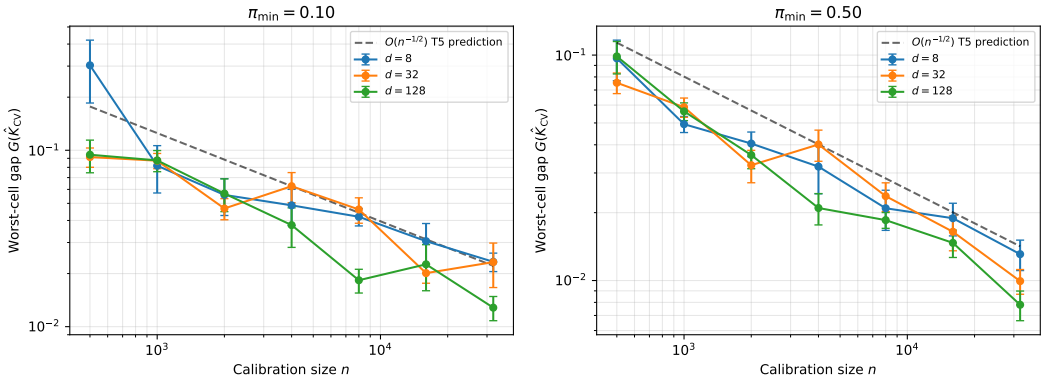

Figure 7: Synthetic n-sweep validating the $O(n^{-1/2})$ rate of Theorem 3. Each panel: empirical worst-cell gap $G(\hat{K}_{\mathrm{CV}})$ vs calibration size n, with ± 1 SE error bars over 5 replicates. Three coloured curves: feature dimensions $d \in \{8, 32, 128\}$. Black dashed: theoretical $O(n^{-1/2})$ slope anchored at $n = 4{,}000$, $d = 32$. The three dimension curves overlap inside each panel (dimension-free rate); the right panel sits below the left by a factor of ≈ 2 at matched n (the $\pi_{\min}^{-1/2}$ constant of T5.1).

graceful-degradation property of T3.b. The empirical drifts are well within T3.b's analytical bound ($\sim 2\eta\bar{\sigma}/\Delta$ would be ≥ 0.5 for $\eta = 0.5$), suggesting the bound is loose but not vacuous in this regime.

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract states the three contributions (T5.1/T5.2 finite-sample rate within the equi-bin Mondrian-K class with $\pi_{\min}^{-1/2}$ explicit; HCCP framework with T3 / T3′ guarantees; empirical validation on TraitGym + ProteinGym + synthetic), and §1 expands these with explicit scope qualifiers ("of the methods we evaluated", "to our knowledge first", "within the equi-bin Mondrian-K class"). The Mendelian $K_{\mathrm{eval}} \geq 5$ regime where HCCP loses to weighted CP is disclosed in the abstract.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: §7 "Assumption discussion" enumerates four limitations (TV proxy approximation, large-sample $K^\star$ versus finite-sample binding, DEGU CNN end-to-end out of scope, ProteinGym 6/50 outliers); §6.4 reports the M→C cross-dataset failure mode case study; §6.3 reports the Skeletal/connective OMIM cluster ($\mathrm{cov}_{|Y=1} = 0.745$) as the worst-cell exception; the $\mathrm{K}_{\mathrm{eval}}$ sensitivity sweep (App. C.1) reverses the H2H story on Mendelian at $K_{\mathrm{eval}} \geq 5$ and is reported transparently.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: All four theorems (T3, T3′, T5.1, T5.2) state assumptions explicitly via the labeled 1/2/3/4 hierarchy. Proofs are in App. A, and the main paper §5 contains proof sketches. Auxiliary lemmas (T1, T2, T1′, T2′, T3-loc, T3.b, T4) and their full proofs are also in App. A.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Datasets (TraitGym matched_9 [Benegas et al., 2025], ProteinGym [Notin et al., 2023]) are public; chrom-LOO splits ($\{17{-}22, X\}$) are TraitGym-canonical; aggregator (HistGradientBoostingClassifier depth 2, 100 iterations, class-balanced), $\hat{\sigma}$ head (same backbone, Gaussian NLL on chrom-LOO OOF residuals), nested chrom-LOO K-selection protocol ($K_{\text{eval}} = 3/5$ keeping ≥ 100 minority per cell), and bootstrap protocol ($B = 200$ chrom-resamples) are all specified in §4, §6, and App. B.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: TraitGym matched_9 and ProteinGym are publicly distributed by their creators. Code (aggregator training, HCCP calibration, nested-CV K-selection, bootstrap) is provided as anonymized supplementary material at submission time and will be released under a permissive licence on acceptance; pipeline scripts (`T_tools/cp_baselines_h2h.py`, `T_tools/bootstrap_cp_baselines.py`, `T_tools/asl_audit.py`) are referenced in App. B.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: §6 setup paragraph specifies splits, features, AUPRC weighting, $\alpha = 0.10$, and K-selection protocol; App. B provides the full HistGB hyperparameters (depth, iterations, class-balanced, default subsample/max_features, random_state inactive at the configuration used), $\hat{\sigma}$-bin edge construction (calibration-fold quantiles), and the small-cell fallback ($n_{k,b} < \lceil 1/\alpha \rceil$ triggers pooled T2 threshold).

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Tab. 2 HCCP rows report $\pm$ one std from $B = 200$ chromosome-level bootstrap resamples; Tab. 17 reports per-method bootstrap mean $\pm$ std plus 95% CIs; Tab. 8 reports point estimates across K_{eval} for sensitivity. Bootstrap captures calibration-set variability (chrom-resample with replacement); single-seed point estimates are flagged as such in Tab. 16 caption.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: App. B (compute table) reports: HCCP pipeline (aggregator + $\hat{\sigma}$ head + Mondrian calibration) runs entirely on CPU 16-core; total ≤ 4 CPU-hour per feature set per dataset; $B = 200$ chrom-bootstrap adds ~ 30 CPU-min total. ProteinGym 50-assay LOO runs on CPU 16-core in < 10 CPU-hour. No GPU is required for the headline pipeline.

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: We have read the NeurIPS Code of Ethics. The paper uses publicly available variant-effect benchmark datasets (TraitGym, ProteinGym) under their respective open licences; no human-subject data collection or crowdsourcing was performed. Clinical-deployment risks of variant prediction are discussed in §7.

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: §7 broader-impact paragraph discusses (i) clinical deployment risks of variant prediction and the need to integrate HCCP singletons into ACMG/AMP multi-evidence frameworks rather than as stand-alone diagnostics; (ii) population representativeness limitations of TraitGym (over-represents European-ancestry individuals from ClinVar / GWAS catalog); (iii) ancestry-stratified Mondrian as a future-work direction for equitable deployment.

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: HCCP is a post-hoc conformal-calibration framework consuming an arbitrary classifier $\hat{p}$ and variance head $\hat{\sigma}$. It is not a pre-trained generative model, language model, or scraped dataset, and we do not release any high-risk asset. All datasets used (TraitGym, ProteinGym) are already public and were created by their original authors with their own release decisions.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: TraitGym [Benegas et al., 2025], ProteinGym [Notin et al., 2023], CADD [Kircher et al., 2014], Borzoi [Linder et al., 2025], GPN-MSA [Benegas et al., 2023], AlphaGenome [Avsec et al., 2026] are all cited; the `crepes` package [Boström, 2022] is used for the H2H baselines and acknowledged. We use each asset in accordance with its public licence; full licence inventory in App. B.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: New assets (HCCP calibration code, A-SL audit utility, bootstrap CP baseline runner) are anonymized and supplied as supplementary material; App. B documents the script entry-points, expected inputs / outputs, and runtime estimates. Final release on acceptance will follow standard open-source documentation templates.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing or research with human subjects. All datasets used are publicly distributed variant-effect benchmarks (TraitGym derived from ClinVar / GWAS catalog; ProteinGym from published deep-mutational-scanning assays).

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The paper does not conduct human-subjects research. Datasets are public benchmarks created and released by their original authors under ethics frameworks of the originating institutions; we use the released aggregate variant labels and do not access any individual-level data.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs were not used as any component of the core methodology. The aggregator $\hat{p}$ is a gradient-boosted classifier and the variance head $\hat{\sigma}$ is a gradient-boosted regressor; both consume pre-computed numeric features (CADD, Borzoi, GPN-MSA scores) rather than language. LLM use was confined to writing assistance.